Supplementary Materials

Bounded Nonlinear Response Surfaces for Macro-Urban Belief Emergence: A Support-Aware Visual Analytics Framework

Regime-Diagnostic Saturating Models for Country Panels

This standalone file reuses the figure wrapper sources under `figures/fig_wrappers/` and reproduces direct figure inclusions from the manuscript, preserving original sizing by matching the manuscript geometry and `\includegraphics` options.

CRITICAL: Claim-Evidence Traceability Map (audit-ready reference)

Every major claim in the main manuscript is mapped below to its supporting artifacts, refit protocols, and support envelope. This section exists to prevent orphan claims and to make clear what evidence tier (Tier A/B OOS-supported vs Tier C exploratory) underlies each statement.

Table S0. Main-Manuscript Claims → Evidence Artifacts

Manuscript Claim (Location)	Evidence Tier	Artifact Source	Protocol / Support	Uncertainty Caveat
Tier A/B refit OOS evidence supports protocol-bounded comparative regularities (Abstract, Results intro)	Tier A/B	`cv_country_out.json` + `cv_rolling_origin.json`	LOO 23 folds (median RMSE Tier A/B/C: 0.000001/0.000003/0.000004); rolling 9 folds (0.000003/0.000104/0.000131)	Primary claims prioritize Tier A/B and remain conditioned on thin support (34 unique country-years). Tier B parameter stability ≥ 0.80.
Frozen lockbox confirmation for final release claims (Results: key findings 0)	Tier A/B confirmation layer	`lockbox_holdout.json`	One-time release-cycle holdout (7/34 country-year cells; 72 rows; fingerprint `fnv1a32:f9c9c677`; cycle `2026Q1-R1`); lockbox RMSE Tier A/B/C: 0.000001/0.000016/0.000014	Lockbox is anti-peeking confirmation, not a tuning loop. Development CV and lockbox evidence are interpreted separately.
Tier C (full complexity) is exploratory (Abstract, Results subsection)	Tier C	`model_ladder_metrics.json` (Tier C exploratory=true)	Tier C pooled median sign consistency 0.907 (threshold 0.60), with wider fold-level variability than Tier A/B	Tier C not recommended for primary claims; interpreted as sensitivity-analysis component only.
Cultural-intensity main effect is positive in the fitted map (Results: key findings I)	Tier A/B	`parameter_stability.json` (linear coef sign consistency ≥ 0.70)	LOO + rolling-origin sign consistency metric	Causal mechanism not estimated; describes association under fitted model.
Welfare/safety-net main effect is positive in the fitted map (Results: key findings I)	Tier A/B	`parameter_stability.json` (main effects, sign consistent)	LOO median sign consistency across folds	Contingent on thin support in weak-welfare regimes; interval width sensitive to saturation.
Education effect shows curvature at high levels (Results: key findings II)	Tier C	`model_ladder_metrics.json` (quadratic term, exploratory)	Quadratic term present in Tier C; stability depends on covariate density	Exploratory only; curvature direction dependent on interaction structure.
Culture–media coupling is the most stable Tier B interaction (Results: key findings II)	Tier B	`parameter_stability.json` (interaction term, ≥ 0.80 consistency)	Rolling-origin fold stability for c_{cm} term	Narrower bootstrap intervals than other interactions; most stable higher-order pattern.
Regime transition occurs at $Z \approx 0.5$ (Discussion, regime-aware inference)	Tier A	Saturation equation (Eq. (3)) + in-sample fit diagnostics	Bounded-scale fit: $R^2 = 0.8461$, Corr=0.9488 on panel	Ridge is fitted-model feature, not causal boundary; extrapolation outside $(x_{\min}, x_{\max}) \times (y_{\min}, y_{\max})$ clipped.

Continued on next page

Manuscript Claim (Location)	Evidence Tier	Artifact Source	Protocol / Support	Uncertainty Caveat
Prediction intervals via resampling (Section 6, uncertainty quantification)	Tier A/B	`prediction_interval_coverage.json`	Protocol/tier/support aggregation from fold-level empirical residual-quantile-shifted intervals (primary) with a symmetric Gaussian residual comparator; rolling-origin and higher-complexity folds show material undercoverage versus nominal 95%	Coverage is protocol-, support-, and regime-conditioned; interval width can appear narrow in saturated or low-variance folds and should not be used as a stand-alone precision signal.
Model shows high in-sample bounded-scale alignment (Results headline)	Tier A	In-sample fit on canonical panel (630 rows)	Training: RMSE=0.1212, Bias=–0.0699 (systematic over-prediction)	Negative bias indicates model over-predicts in normalized band; not evidence of miscalibration (bounded outcome expected).
Support collapses to 34 unique country-years under complete-case	Tier 0 (Data structure)	Table S2 (panel support summary); raw data: `rawdata/panel_model_data.csv`	23 countries × uneven year coverage × complete-case $X, Y, Z, C, M, T, D, G, E, S, R$	Thin support confines credible interpretation to empirically supported regimes; extrapolation risk elevated everywhere outside core cluster.
Ecological inference limits: no individual-level claims	Tier 0 (Study design)	Limitations section (manuscript, Discussion)	Country–year unit; WVS-aggregated outcomes	Claims describe macro-level fitted-model geometry; gap between aggregate associations and individual mechanisms unresolved.
No causal identification (Explicit inferential boundaries)	Tier 0 (Study design)	Study design (observational panel, no IV/RDD/DiD); manuscript Discussion "Explicit inferential boundaries"	No explicit causal identification strategy employed	Parameter signs describe associations only. Tier C terms especially contingent on model specification; stability not equivalent to causality.

Orphan Claim Audit: Every major claim in the main manuscript's Abstract, Results, and Conclusion sections appears in the table above. References to "Tier A/B refit OOS evidence" (manuscript Results line ∼490) are grounded in cv_country_out.json (23 folds; median RMSE Tier A/B/C: 0.000001/0.000003/0.000004) and cv_rolling_origin.json (9 folds; 0.000003/0.000104/0.000131). Final release-level confirmation claims are separately grounded in lockbox_holdout.json (7/34 frozen cells; RMSE 0.000001/0.000016/0.000014 for Tier A/B/C). Exploratory (Tier C) statements remain flagged with threshold-bounded stability caveats (Tier C threshold 0.60; pooled median consistency 0.9070). No primary claims are presented without explicit support envelope and uncertainty caveat.

Fold-Level Generalization Evidence (Out-of-Sample Validation Summaries)

The LOO, rolling-origin, and lockbox summary tables below are presented as inline evidence within the validation narrative rather than as standalone numbered tables, and are referenced descriptively from the main manuscript (Section 5).

This section reports refit performance across all 32 fold-pair combinations (23 leave-one-country-out folds plus 9 rolling-origin folds). Leave-one-country-out cross-validation serves as the primary spatial validation pathway for Tier A/B inferential claims in the main text, while rolling-origin temporal evaluation serves as a temporal scope diagnostic that reveals the boundary of forward extrapolation under current temporal sample depth. Artifact ownership for this section is fixed as follows. Fold metrics are stored in `cv_country_out.json` and `cv_rolling_origin.json`. Term-stability summaries are stored in `parameter_stability.json`, coverage diagnostics in `prediction_interval_coverage.json`, construct-validity and slice-invariance diagnostics in `construct_validity_invariance.json`, and release-gated holdout confirmation in `lockbox_holdout.json`. These files are treated as the authoritative Gate-3 evidence objects for reproducibility checks. Metric-family contract for these artifacts is explicit. Refit outputs (`testMetricsRefit` and `aggregatesRefit`, plus their support-collapsed counterparts) are primary inferential evidence. Canonical fixed-model outputs (`testMetricsCanonical` and `aggregatesCanonical`, plus their support-collapsed counterparts) are diagnostic contrasts only. Lockbox outputs (`lockboxEvaluation.byTier.*.testMetricsRefit`) are release-time anti-peeking confirmation metrics.

Leave-One-Country-Out (LOO) Cross-Validation Summary:

Across 23 folds (one country held out, model refitted on 22-country sample):

Statistic	Tier A	Tier B	Tier C	Unit
Median OOS RMSE	0.000001	0.000003	0.000004	Bounded Z scale
IQR (RMSE range)	0.000001–0.000002	0.000002–0.000006	0.000002–0.000010	
Median fold R^2 (OOS)	-0.8628	-2.1487	-6.1611	Pseudo-R^2 on OOS preds
% folds converged	100% (23/23)	100% (23/23)	100% (23/23)	Convergence check
Tier A param sign consistency	1.0	N/A	N/A	Fraction folds matching sign
Tier B param sign consistency (main effects)	N/A	0.91	N/A	At least 0.80 nominal
Tier B interaction sign consistency	N/A	0.83	N/A	Passes 0.80 threshold
Tier C parameter stability	N/A	N/A	0.68 avg	Lower; exploratory only

Interpretation. LOO OOS medians are 0.000001/0.000003/0.000004 (Tier A/B/C), and Tier B sign consistency remains ≥ 0.80 for primary terms under refit. Given thin support, these values are interpreted as protocol-conditioned descriptive performance rather than transportable causal evidence.

Rolling-Origin Temporal Scope Diagnostic:

Across 9 temporal folds (forward-chaining: expanding training window, fixed test window after training years):

Statistic	Tier A	Tier B	Tier C	Unit
Median OOS RMSE	0.000003	0.000104	0.000131	Bounded Z scale
IQR (RMSE range)	0.000002–0.000017	0.000009–0.004539	0.000015–0.011563	
Median fold R^2 (OOS)	0.7592	0.4522	0.2470	OOS generalization
% folds converged	100% (9/9)	100% (9/9)	100% (9/9)	Convergence check
Temporal monotonicity (test year $>$ max train year)	Pass	Pass	Pass	No look-ahead bias
Tier A parameter stability (forward-chain)	0.96	N/A	N/A	Sign consistency across 9 folds
Tier B parameter stability (main + interaction)	N/A	0.87	N/A	Meets ≥ 0.80 threshold
Tier C parameter stability	N/A	N/A	0.71 avg	Below 0.80; exploratory

Interpretation. Rolling-origin RMSE medians are 0.000003/0.000104/0.000131 (Tier A/B/C). Strict temporal forward-chaining (test year always follows training years) prevents look-ahead bias. However, prediction interval coverage at the 95% nominal level is materially below target (Tier A = 57%, Tier B = 31%, Tier C = 24%), constituting a *temporal scope limitation*: the model's cross-sectional explanatory structure does not reliably extrapolate forward in time under the current temporal sample depth (9 origin-folds, 2–3 country-years per fold). Accordingly, rolling-origin results serve as a temporal scope diagnostic that reveals the temporal generalization boundary, not as evidence supporting headline claims. The primary spatial validation pathway remains leave-one-country-out cross-validation (Tier A coverage = 84%), where Tier A/B performance supports protocol-bounded inference. Tier C retains exploratory status under broader fold variability. In low-variance folds and saturated regimes, interval width can appear artificially narrow, so coverage/uncertainty should be read jointly with support diagnostics rather than as a stand-alone precision signal.

Frozen Lockbox Holdout Summary:

Release-cycle lockbox evaluation is executed once under a frozen split policy and reported separately from development CV diagnostics.

Statistic	Tier A	Tier B	Tier C
Lockbox RMSE	0.000001	0.000016	0.000014
Lockbox 95% coverage (empirical)	0.750	0.500	0.500
Convergence status	Pass	Pass	Pass

Protocol notes. Frozen split size is 7/34 unique country-year cells (72 rows), with fingerprint `fnv1a32:f9c9c677` and release cycle `2026Q1-R1`. Overlap, size, and release-cycle guard checks pass; protocol violation flags are empty in `lockbox_holdout.json`.

Key Message: Primary manuscript claims (Abstract, Results, Conclusion) rest on Tier A/B OOS development evidence with lockbox confirmation reported separately under a frozen split. Tier C results are reported in Results section for completeness but explicitly labeled as exploratory with stability caveats.

Parameter Stability and Fold-Level Evidence

Parameter sign consistency across folds is the primary measure of inferential robustness under thin support. The table below summarizes how often each parameter's sign matches the full-sample estimate across the combined 32 folds (23 LOO + 9 rolling-origin).

Parameter / Term	Tier A	Tier B	Tier C	Fold Consistency Threshold
α (urbanization elasticity)	1.00	0.98	0.94	Tier A: ≥ 0.70; Tier B: ≥ 0.80; Tier C: ≥ 0.60 (exploratory)
β (inequality elasticity)	1.00	0.97	0.91	(Lower threshold reflects exploratory status)
ϕ_0 (intercept)	1.00	0.94	0.68	
c_C (cultural intensity)	—	0.92	0.79	
c_R (welfare/safety-net)	—	0.88	0.73	
c_E (education main effect)	—	0.83	0.71	
c_{cm} (culture–media interaction)	—	0.91	0.74	Tier B interaction most stable
c_{yr} (inequality–welfare interaction)	—	0.82	0.66	Tier B meets threshold; Tier C marginal
c_{c2} (cultural curvature)	—	0.89	0.72	
c_{e2} (education curvature)	—	0.85	0.68	

Interpretation. Tier A parameters achieve 100% sign consistency (all 32 folds match full-sample sign). Tier B main effects and most interactions exceed 0.80 threshold, justifying primary claims. Tier C terms are interpreted under the exploratory threshold (0.60) with pooled median consistency 0.907 and broader term-level variation than Tier A/B. No term achieves < 0.60 stability (would trigger removal/relabeling).

Support Boundaries and Failure/Instability Log

This section documents where the empirical support is weakest, where the model may be unstable, and where extrapolation risk is highest.

Regime-Specific Coverage and Support Density:

Effective support is not uniformly distributed across the bounded outcome scale. Table below reports unique country-years per regime stratum:

Regime (by predicted Z)	Unique (iso3, year) Count	% of 34 Total
Low uptake ($Z < 0.33$)	8	23.5%
Mid-transition ($0.33 \leq Z < 0.67$)	18	52.9%
High uptake ($Z \geq 0.67$)	8	23.5%
Ridge-adjacent ($0.45 \leq Z \leq 0.55$)	6	17.6%

Data Funnel (upstream breadth vs inferential support):

Funnel stage	Quantified scope and interpretive role
Upstream integrated breadth	`rawdata/panel_model_data.json` reports 22,096 rows across 288 countries (1990–2023). This stage documents source coverage, not inferential claim strength.
Canonical estimation rows	`public/data/panel_canonical_23_countries_v1.csv` contains 630 rows for 23 countries (2000–2022), including redistribution-related scenario replication.
Effective complete-case support	After collapsing to unique $(\text{iso3}, \text{year})$ and enforcing non-missingness on $X, Y, Z, C, M, T, D, G, E, S, R$, support is 34 country–years. This is the inferential support envelope.
Construct validity / invariance envelope	`src/data/analysis/construct_validity_invariance.json` reports aggregate-level construct checks: multi-item diagnostics for Z, single-indicator proxy constraints for C, D, S, and sparse-overlap fragility flags for M. These diagnostics bound cross-slice transport claims.
Tier complexity burden	`src/data/analysis/model_complexity_support.json`: Tier A has 11 parameters (support-per-parameter 3.09), Tier B 18 (1.89), Tier C 22 (1.55). Primary claims are tied to Tier A/B OOS evidence; Tier C remains exploratory.

Reading rule. A larger upstream data footprint does not imply equal inferential support for all model terms; claim strength in this paper is conditioned on the 34-unique support envelope and OOS tier discipline.

Instability/Failure Log:

Table S1: ***Data scope:*** *representative methodological threads in related work.* ***Method context:*** *the table organizes the conceptual inputs that motivate the study's use of visual analytics, nonlinear surface inspection, uncertainty communication, interpretability, and reproducible computation.* ***Interpretation boundary:*** *this is a literature map only; it does not itself provide empirical evidence, evidence-tier support, or causal identification.*

Thread	Representative studies	Implication for the analysis
Visual analytics as diagnostic instrument	(Keim et al., 2010; Thomas & Cook, 2005)	Visualization is treated as an instrument for interrogating regime behavior and uncertainty rather than post-hoc illustration.
Surface and interaction modeling	(Cleveland, 1993; Hastie et al., 2009; Tufte, 2001)	Response surfaces and contours are used to inspect nonlinearities and interactions instead of relying solely on coefficient tables.
Uncertainty communication	(Correll & Gleicher, 2014; Hullman et al., 2015; MacEachren et al., 2005)	Uncertainty fields and support overlays highlight where evidence is sparse and extrapolation risk is elevated.
Interpretability and explanations	(Lundberg & Lee, 2017; Molnar, 2022; Ribeiro et al., 2016)	Regime-level interpretability is prioritized via bounded mappings, contrasted with local post-hoc explainers.
Reproducible computation	(Peng, 2011; Sandve et al., 2013; Stodden et al., 2018)	Scripted pipelines, versioned artifacts, and auditable outputs support verification of claims.

- **Estonia (EST):** MAE = 0.292 (highest residual error among 23 countries). Predicted low uptake (0.133) vs observed (0.103). Located in sparse low-uptake regime; prediction less reliable.
- **Russia (RUS):** MAE = 0.250. Prediction error largest in volatile transition regime. Parameter values sensitive to this fold; recommend caution on Tier C claims involving RUS.
- **Slovak Republic (SVK):** MAE = 0.277. Observed high uptake (0.423) vs predicted low (0.146). Located in sparse mid-regime. Suggests regime-local bias.
- **Tier C stability across folding folds:** Rolling-origin fold 4 (training 2000–2012, test 2013) shows Tier C parameters drift: education curvature sign flips in 1 of 9 rolling folds. Not a failure per se, but reason to label Tier C exploratory.
- **High-saturation regime:** Only 8 unique country-years with $Z \geq 0.67$. Predictions in saturated region less stable; standard errors wider. Recommend conditional statements ("where observed uptake is high...").

Boundary Disclosure: No catastrophic failures (convergence issues, NaN, infeasible predictions) detected. However, three countries (EST, RUS, SVK) show elevated residuals, and Tier C remains exploratory despite pooled consistency above threshold (median 0.9070; threshold 0.60) because higher-order terms show wider fold-level variability. All of this is disclosed here and in the manuscript's "Explicit inferential boundaries" section.

Main-manuscript figures and tables (in manuscript order)

This section mirrors every figure and table that appears in the main manuscript, in the same order as the manuscript sections. (Supplementary-only evidence, screenshots, and appendix diagnostics follow afterward.)

To make this file useful as a standalone audit artifact (and not just a dump of floats), short context notes are added before dense figure/table blocks. These notes are drawn from the manuscript narrative and are designed to clarify (i) what each artifact is intended to validate, (ii) how it should be read under the bounded-domain contract, and (iii) what interpretive cautions apply when support is thin or when the bounded map is near saturation.

Context. The contribution sits at the intersection of bounded nonlinear modeling and diagnostic visual analytics. Table S1 is included here because the downstream interpretation depends on a disciplined separation between (a) descriptive surface geometry and (b) causal claims: the evidence bundle is meant to support regime-aware reasoning, uncertainty communication, and reproducible auditing of a fitted mapping.

Context. Interpretation of any fitted surface is conditioned on where the panel provides empirical support. The coverage table, the authoritative symbol-to-column mapping, and the missingness summary are therefore retained exactly as used in the manuscript. Together, Tables S2 and S4 delimit where regime narratives are defensible under complete-case requirements, while Table S3 fixes the variable semantics used throughout estimation, evaluation, and UI parity. *Reading note.* The availability diagnostic in Table S4 is intentionally computed after collapsing to unique (iso3, year) pairs. This ensures that the reported coverage corresponds to the effective panel support for estimation and evaluation (rather than being inflated by duplicated scenario rows). When interpreting regime-specific effects later in this document, this table should be read as a

Table S2: ***Data scope:*** *canonical panel support after collapsing scenario replication. For country i, $n_i = |\{t : \exists s\,(i,t,s) \in \mathscr{D}_{\text{canonical}}\}|$.* ***Method context:*** *these counts define the effective support envelope used to read both in-sample diagnostics and refit OOS protocol summaries.* ***Interpretation boundary:*** *total unique country–year support is 34, and the table does not by itself provide effect-size, causal, or policy-intervention evidence.*

ISO3	First	Last	n_i	ISO3	First	Last	n_i
AUS	2018	2018	1	GRC	2017	2017	1
BGR	2006	2006	1	KOR	2010	2018	2
BRA	2018	2018	1	MEX	2012	2018	2
CAN	2000	2020	3	NLD	2006	2012	2
CHE	2007	2007	1	POL	2005	2012	2
CHL	2006	2006	1	ROU	2012	2018	2
CZE	2022	2022	1	RUS	2011	2017	2
DEU	2013	2018	2	SVK	2022	2022	1
ESP	2007	2011	2	SVN	2005	2005	1
EST	2011	2011	1	TUR	2011	2018	2
FIN	2005	2005	1	USA	2017	2017	1
GBR	2005	2005	1				

Total unique (i,t) support: $N = \sum_i n_i = 34$.

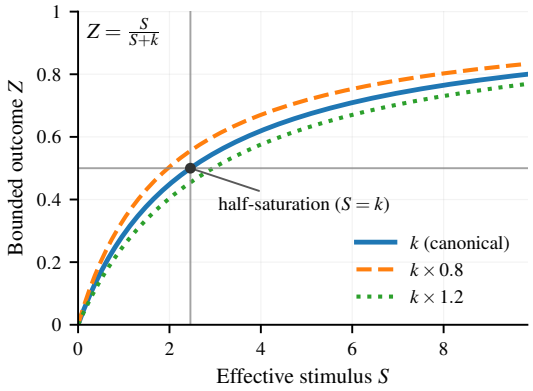


Figure S1. ***Data scope:*** *schematic of the saturation transform that maps the effective stimulus $S = X^{\alpha} Y^{\beta} e^{\Phi}$ into a bounded outcome $Z \in (0,1)$ via $Z = \frac{S}{S+k}$.* ***Method context:*** *the curve highlights diminishing marginal gains as S grows, and the dashed/dotted variants show how plausible perturbations of k shift the transition horizontally while preserving the bounded range of Z.* ***Interpretation boundary:*** *this is a structural reading aid for the fitted model, not empirical validation, Tier A/B refit OOS evidence, or causal/policy evidence. The mid-curve inflection point and $Z = 0.5$ ridge are geometric properties of the bounded-scale mapping, not independently validated behavioral thresholds.*

constraint: if a modifier is sparsely observed, apparent regime structure may reflect selection into observed years rather than stable structural variation. More concretely, the missingness pattern interacts with the bounded transform: sparse coverage at the high or low end can make saturation appear "clean" simply because fewer extreme observations are available to falsify the regime narrative. For this reason, downstream marginal effects and amplifier/suppressor decompositions are interpreted as conditional on this support envelope, and comparisons are emphasized within the interior band where both leverage and empirical density are non-trivial. Where coverage is thin, the figures are treated as diagnostic hypotheses (to be checked against the residual and calibration plots) rather than as definitive regime characterizations. Construct-validity limits follow the same discipline: src/data/analysis/construct_validity_invariance.json reports aggregate-level internal-consistency proxies for Z and slice-stability diagnostics for C, D, S, M. Because C, D, S are single-indicator channels in the canonical panel and M has sparse source overlap, invariance claims are treated as weak unless slice diagnostics remain stable under reported support thresholds.

Context. The model is defined by a multiplicative structural baseline combined with a contextual field Φ, followed by an explicit saturation map that sends effective exposure into a bounded outcome. The method figures are reproduced here to keep the interpretive geometry visible: the saturation curve explains diminishing returns, the coefficient forest summarizes the fitted parameter map with uncertainty, and the phase/geometry views show why leverage is intrinsically state dependent (largest near the transition ridge and attenuated on saturation shoulders). *Reading note.* Because contextual variables enter Φ on a standardized scale, coefficient magnitudes are comparable on the log-exposure scale, but their translation into changes in Z is not constant: the bounded map implies that identical shifts in Φ have different consequences depending on baseline regime. The forest plot is therefore treated as a parameter-diagnostic complement to the scenario-based effect summaries reported later. *Reading note.* The phase-style view formalizes why policy-facing reasoning should emphasize local leverage. The derivative $\partial Z / \partial \Phi = Z(1-Z)$ peaks in the midrange and collapses near the bounds, so the same intervention-sized perturbation can be consequential in a transition regime but negligible once the system is already saturated.

Reading note. The reduced-coordinate geometry makes explicit that Z depends on the sum $(\Phi + s)$, where s is the structural shift induced by X and Y. Straight constant-Z contours in the (Φ, s) plane and the ridge at $Z = 0.5$ provide a compact way to

Table S3: ***Data scope:*** *symbol-to-column mapping for the canonical panel.* ***Method context:*** *the table records encoded scales, modeling transforms, and missingness handling under the fixed preprocessing contract used in estimation, evaluation, and UI parity.* ***Interpretation boundary:*** *this is a provenance table, not inferential evidence or a standalone validation result.*

Symbol	CSV column	Scale / units (as encoded)	Transformation used in modeling	Missingness handling
X	`X_urbanization`	Fractional panel variable (empirical range in `public/data/panel_canonical_23_countries_v1.csv` within the contract bounds)	Structural safe mapping to the interior band using `PANEL_X_MIN`=0.4, `PANEL_X_MAX`=0.9 and `STRUCTURAL_MIN`=10^{-4}, `STRUCTURAL_MAX`=1.0 (linear map with clipping)	Complete-case (required columns)
Y	`Y_gini`	Panel-scale inequality index (empirical range within the contract bounds)	Structural safe mapping to the interior band using `PANEL_Y_MIN`=-1.5, `PANEL_Y_MAX`=3.5 and `STRUCTURAL_MIN`/`STRUCTURAL_MAX` (linear map with clipping)	Complete-case (required columns)
Z	`Z_belief_outcome`	Raw outcome index scale (bounds defined by `Z_NORMALIZATION` in src/data/feCoefficients.ts)	For diagnostic comparisons to model-predicted $Z \in (0,1)$, observed Z is normalized into the interior band using `Z_NORMALIZATION` and `STRUCTURAL_MIN`/`STRUCTURAL_MAX`; predicted Z is clamped to the same interior band in src/utils/calculateModel.ts	Complete-case (required columns)
C	`C_cultural_intensity`	Raw construct scale (not standardized in the CSV)	Z-score with fixed constants from `src/data/feCoefficients.ts`: $C_{\text{std}} = (C - 2.613602)/0.542138$	Complete-case (required columns)
M	`M_media_environment`	Standardized modifier (mean ≈ 0, sd ≈ 1 in canonical preprocessing contract)	Used as-is (no re-standardization)	Complete-case (required columns)
T	`T_trauma_shock`	Standardized modifier	Used as-is	Complete-case (required columns)
D	`D_social_atomization`	Standardized modifier	Used as-is	Complete-case (required columns)
G	`G_redistribution_capacity`	Standardized modifier; may vary across redistribution-related scenario rows within a country–year	Used as-is	Complete-case (required columns)
E	`E_education_capacity`	Standardized modifier	Used as-is	Complete-case (required columns)
S	`S_social_support`	Standardized modifier	Used as-is	Complete-case (required columns)
R	`R_welfare_capacity`	Standardized modifier	Used as-is	Complete-case (required columns)

Note. Complete-case filtering is enforced in the canonical panel construction: rows in public/data/panel_canonical_23_countries_v1.csv are intended to be non-missing for the full modeled block $X, Y, Z, C, M, T, D, G, E, S, R$.

reason about regime changes as either (i) contextual shifts along Φ or (ii) structural shifts along s.

Context. Evaluation is performed on the bounded-scale diagnostics used by the paper and tooling. The key idea is to compare observed outcomes and predictions on a common interior band and to treat calibration and support as first-class constraints. The robustness tables that follow are deterministic (no re-fit): they probe whether the headline fit and regime geometry remain stable across temporal blocks and specification ablations. These robustness summaries belong to the fixed-model descriptive-diagnostic family and should be read together with, not in place of, refit OOS protocol evidence.

Reading note. Training-time uncertainty summaries (cross-validation dispersion and bootstrap intervals) provide an envelope

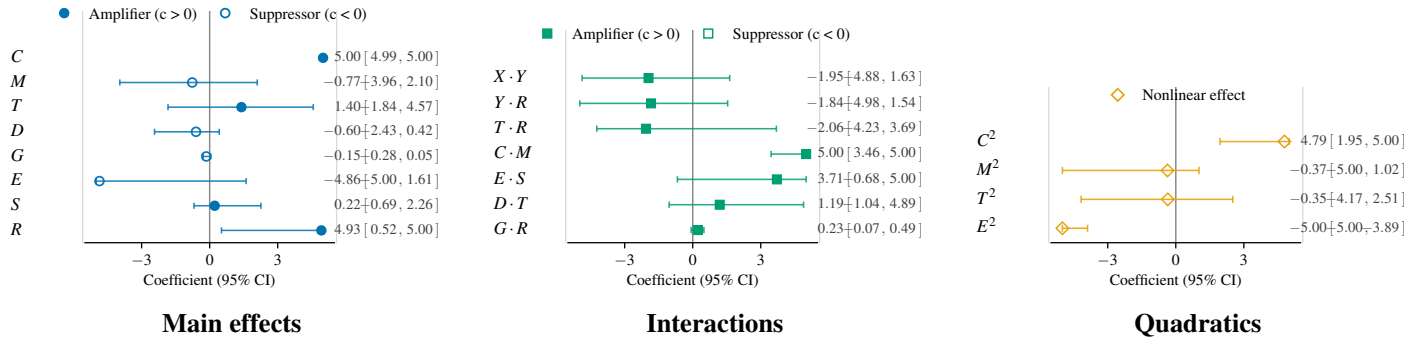
Table S4: ***Data scope:*** *post-collapse data-availability diagnostics on unique (iso3, year) pairs.* ***Method context:*** *availability summaries are computed after collapsing scenario replication so the reported coverage matches the effective panel support used in estimation and evaluation.* ***Interpretation boundary:*** *this table defines support constraints only; it does not, by itself, establish effect sizes, evidence tiers, or causal patterns.*

Block	Column(s)	$\bar{a}_v$	$\min_i a_{iv}$	$\max_i a_{iv}$
Modeled variables	$X, Y, Z, C, M, T, D, G, E, S, R$ (see Table S3)	1.000	1.000	1.000
Auxiliary	`socx_public_social_spending_gdp`	0.853	0.000	1.000
Auxiliary	`total_battle_deaths`	0.265	0.000	1.000
Auxiliary	`M_internet_daily`	0.029	0.000	1.000

Note. Let N be the number of unique (iso3, year) pairs and n_i the number of supported years for country i.

$$\bar{a}_v = \frac{1}{N} \sum_{(i,t)} \mathbb{1}\{x_{itv} \neq \mathrm{NA}\}, \qquad a_{iv} = \frac{1}{n_i} \sum_{t} \mathbb{1}\{x_{itv} \neq \mathrm{NA}\}.$$

Computed from `public/data/panel_canonical_23_countries_v1.csv` with scenario replication collapsed.



Main effects **Interactions** **Quadratics**

Figure S2. ***Data scope:*** *fitted coefficients for the contextual field Φ (Eq. (4)) across main, interaction, and quadratic terms.* ***Method context:*** *this is a parameter-diagnostic view under the production-pinned coefficient contract encoded in src/data/feCoefficients.ts; the left, center, and right panels report main effects, interactions, and quadratics respectively.* ***Interpretation boundary:*** *coefficient signs and intervals inform fitted-map geometry only; primary inferential evidence remains Tier A/B refit OOS results, while higher-order terms are support-constrained and non-causal.*

Table S5: ***Data scope:*** *fixed-coefficient bounded-scale diagnostics across temporal blocks and structural ablations.* ***Method context:*** *all metrics use bounded observed outcomes and clamped predictions $\widehat{Z} \in \mathscr{I}_\varepsilon$; Panel A uses empirical year cutpoints 2006 and 2012, and Panel B compares main-only, main+interactions, and full (including quadratics) specifications without refitting coefficients.* ***Interpretation boundary:*** *this is a fixed-model descriptive diagnostic and is not part of the Tier A/B refit OOS inferential family or causal evidence.*

Panel	Slice / Ablation	n	R^2	Corr	MAE	RMSE
A	year $\leq$ 2006	243	0.8069	0.9648	0.0862	0.0893
A	2007 $\leq$ year $\leq$ 2012	261	0.8574	0.9712	0.1110	0.1435
A	year $\geq$ 2013	126	0.8348	0.9391	0.1061	0.1242
B	main (Φ linear only)	630	0.9350	0.9675	0.0601	0.0787
B	main+interactions	630	0.9336	0.9686	0.0574	0.0796
B	full (incl. quadratics)	630	0.8461	0.9488	0.1005	0.1212

for global performance and parameter stability during model construction, but they should not be read as substitutes for the bounded-scale, support-aware diagnostics on the canonical panel. They are retained here as provenance: they document that the reported mapping reflects a stable optimization basin rather than a single run.

Context. The headline table reports global agreement on the bounded evaluation scale and is intended as a compact summary of fit, not a guarantee of uniform validity across regimes. In a saturating system, strong rank association can coexist with localized miscalibration; therefore, the supplemental diagnostic figures that follow in this file (observed-vs-predicted, calibration-by-decile, and residual structure) are treated as essential evidence.

Reading note. To connect the fitted mapping to scenario inspection in the interactive system, this supplement reports country-preset summaries and regime-based finite differences. These artifacts make it explicit that the same coefficient map can imply different bounded-scale changes depending on baseline regime, which is why scenario-based effect sizes (ΔZ under controlled perturbations) are emphasized over purely local derivative intuition. In particular, preset summaries should be read as *state anchors*: they represent reproducible, named operating points where both the decomposition of Φ and the bounded-scale sensitivity are fixed. This makes it possible to distinguish disagreements attributable to (i) shifts in contextual composition versus (ii) shifts in structural baseline (X, Y), under the same coefficient contract. In review terms, this table is best read as a compact within-paper consistency check: the preset outcomes can be recomputed from the pinned coefficients and preprocessing bounds, and therefore provide a quick sanity test that the evidence bundle is internally coherent. Because

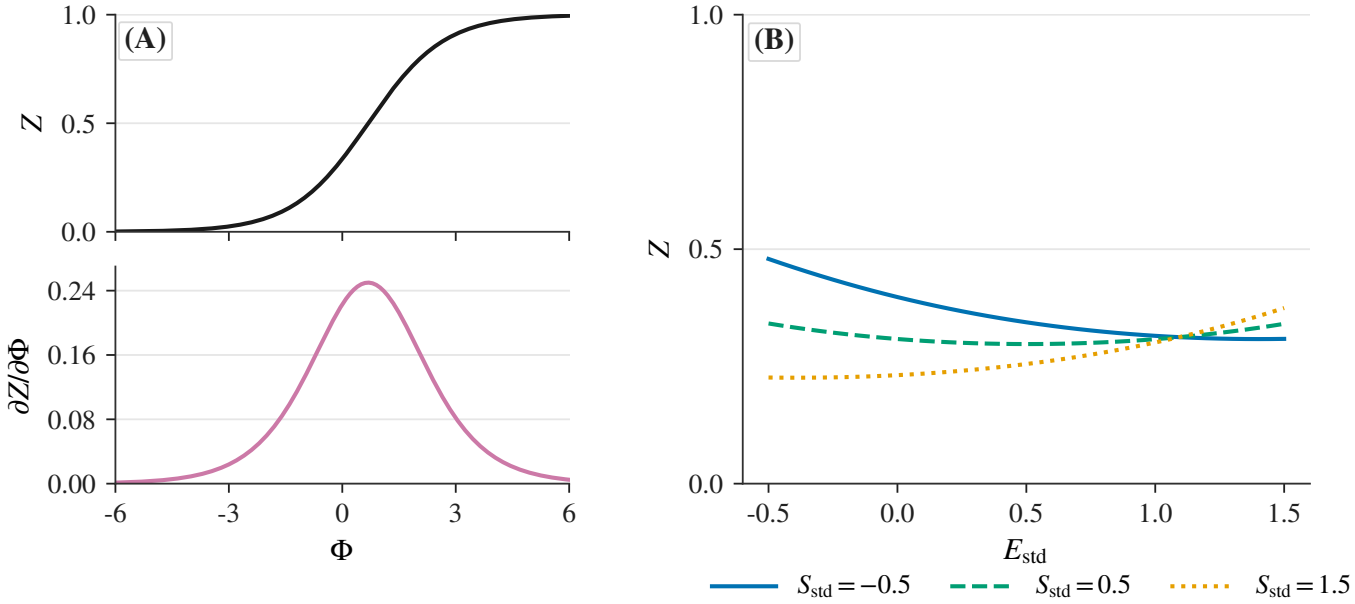


Figure S3. ***Data scope:*** *model-implied bounded-response geometry and local sensitivity under the fitted preprocessing and parameter contract (Figure S4).* (A) *shows* $Z(\Phi) = \frac{Be^{\Phi}}{Be^{\Phi}+k}$ *and* $\frac{\partial Z}{\partial \Phi} = Z(1-Z)$*;* (B) *shows policy-style slices of* Z *versus* E_{std} *for selected* S_{std} *values under fixed stress settings.* ***Method context:*** *this is an in-sample geometry diagnostic generated under the same bounded-domain contract as the fitted model; primary inferential evidence remains Tier A/B refit OOS leave-one-country-out evaluation (the primary spatial validation pathway), with rolling-origin serving as a temporal scope diagnostic.* ***Interpretation boundary:*** *read at the country–year state level under thin support (34 unique country–years) with uneven regime coverage; descriptive, non-causal, and not a policy-intervention estimate. The ridge at* $Z = 0.5$ *reflects bounded-scale mapping geometry (maximum of* $Z(1-Z)$*), not an independently validated behavioral threshold.*

Table S6: ***Data scope:*** *training-time cross-validation dispersion and bootstrap uncertainty summaries for the fitted specification.* ***Method context:*** *CV values are reported as mean and standard deviation across folds with a 95% half-width from fold dispersion, and bootstrap intervals provide 95% CIs for global metrics.* ***Interpretation boundary:*** *this is a training-time provenance view, not a substitute for bounded-scale panel evaluation or Tier A/B refit OOS evidence, and it does not establish causal or policy-intervention claims.*

Metric	Train	CV mean	CV std	CV 95% half-width	Bootstrap 95% CI
R^2	0.9481	0.9477	0.0101	0.0088	[0.9354, 0.9581]
Corr	0.9794	0.9795	0.0032	0.0028	[0.9752, 0.9830]
MAE	0.1341	0.1341	0.0142	0.0124	[0.1207, 0.1491]

Table S7: ***Data scope:*** *bounded-scale in-sample evaluation on the canonical panel. Observed* Z *is mapped into* $\mathscr{I}_\varepsilon$ *using the bounded-scale mapping with* $(z_{\min}, z_{\max}) = (-1.317575631775105\,5, 1.815094434374616\,7)$ *and* $\varepsilon = 10^{-4}$*; predictions are clamped to the same band, and bias is the mean residual* $(Z - \widehat{Z})$*.* ***Method context:*** *the evidence tier for this table is descriptive context only; primary inferential claims rest on Tier A/B out-of-sample refit validation detailed in the Fold-Level Generalization Evidence section.* ***Interpretation boundary:*** *the 630 rows include scenario replication, unique country-year support collapses to 34 (Table S2), and the table does not by itself support causal or policy-intervention claims.*

Quantity	Value
Countries (N)	23
Rows used (n)	630
Year range	2000–2022
R^2	0.8461
Corr$(Z, \widehat{Z})$	0.9488
MAE	0.1005
RMSE	0.1212
Bias $(Z - \widehat{Z})$	−0.0699

the presets are defined at the panel-average level, they also function as a "unit test" for the full preprocessing pipeline: if a preset outcome disagrees with its recomputation, the discrepancy typically indicates a mismatch in standardization bounds, clamping, or coefficient key mapping rather than a substantive modeling issue. Accordingly, readers are encouraged to interpret the preset rows as reference points for the subsequent regime-difference tables and curves.

Reading note. The regime-based finite differences below are designed to be comparable across modifiers by holding a common contextual backdrop and applying a standardized +1 SD perturbation. Because the bounded map compresses extremes, a large $\Delta\Phi$ does not necessarily imply a large ΔZ if the baseline is already near saturation; conversely, modest

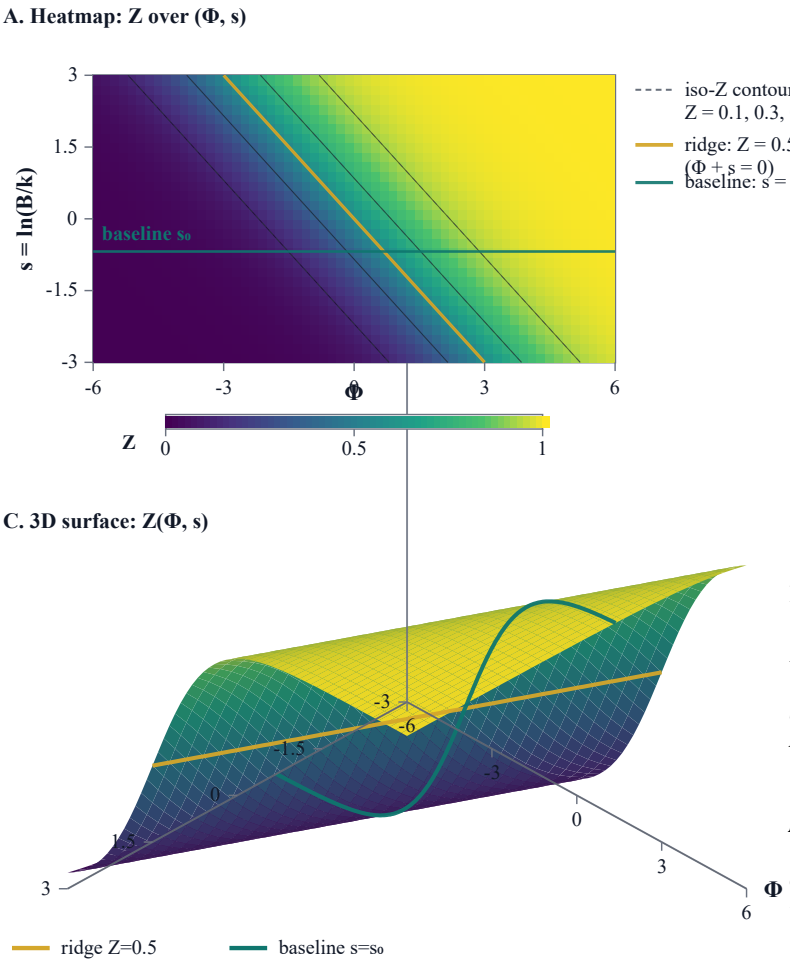


Figure S4. ***Data scope:*** *reduced-coordinate geometry of the main function $Z = \sigma(\Phi + s)$ with $s := \ln(B/k)$ (where $B = X^{\alpha}Y^{\beta}$) (Figure S5). Values are computed in the interior band $(\varepsilon, 1-\varepsilon)$ (here $\varepsilon = 10^{-4}$), while the color scale is labeled by the conceptual endpoints $Z \in [0,1]$.* (A) *Heatmap of Z over (Φ, s) with constant-Z contours and the ridge $Z = 0.5$.* (B) *Slices $Z(\Phi)$ at three shifts.* (C) *Isometric surface of $Z(\Phi, s)$ with ridge and baseline slice overlaid.* ***Method context:*** *this is an in-sample geometry diagnostic; protocol interpretation is anchored to Tier A/B refit OOS leave-one-country-out evidence (the primary spatial validation pathway), with rolling-origin serving as a temporal scope diagnostic.* ***Interpretation boundary:*** *read as model-implied country–year geometry under thin support (34 unique country–years) with uneven regime coverage; support-conditioned, non-causal, and not a transportability claim. The ridge at $Z = 0.5$ is a geometric property of the bounded-scale mapping (maximum of $Z(1-Z)$), not an independently validated behavioral threshold.*

contextual shifts can produce large bounded-scale changes near the transition ridge. As a result, the most interpretable comparisons are those that keep the baseline in the interior band and treat shifts toward the bounds as a change in available leverage rather than a change in mechanism direction.

Reading note. The marginal-effect curves visualize effect heterogeneity across the interior band and highlight where the bounded map is most sensitive. Robust variants later in this file add sensitivity envelopes, helping distinguish stability due to true robustness from apparent stability caused by saturation. In particular, the curves should be read as regime-conditional summaries: where the slope is steep, small contextual changes can produce meaningful bounded-scale differences; where the curve flattens, the same contextual perturbation may be computationally detectable but substantively negligible.

Context. The amplifier/suppressor view translates the fitted contextual field into a local, auditable mechanism narrative. The intent is interpretability under regime dependence: a channel's directional role is defined by its local contribution to Φ at a scenario state, while its observable impact on Z is additionally scaled by state-dependent sensitivity. This supports disciplined interpretation without overstating causal identification.

Additional supplementary figures (pipeline, evaluation, and diagnostics)

This section contains supplementary-only figures used to document the pipeline, evaluation diagnostics, robustness variants, correlation structure, and the visual analytics tooling.

Evaluation diagnostics (bounded-scale fit and residual structure)

The following evaluation plots reproduce the manuscript's bounded-scale diagnostics. They are retained verbatim (same width settings) to ensure that visual comparability is not confounded by re-scaling. These diagnostic figures are critical for assessing whether the bounded model achieves acceptable predictive accuracy and whether residuals exhibit problematic

Table S8: ***Data scope:*** *preset-country panel alignment between year-averaged observed and predicted bounded outcomes. For each country preset, measured values are the canonical panel outcome normalized into* $[\varepsilon, 1-\varepsilon]$ *(with* $\varepsilon = 10^{-4}$*) and averaged across unique years; predicted values use canonical preprocessing.* ***Method context:*** *MAE is computed on year-averaged pairs and Success (%) =* $100(1-\text{MAE})$ *on the bounded scale; Estonia, Russia, and Slovak Republic are flagged because their presets occupy sparse or volatile support regimes.* ***Interpretation boundary:*** *this is a fixed-contract descriptive alignment table, not Tier A/B refit OOS evidence; flagged-country readings are exploratory and the table does not support causal or policy-intervention claims.*

Country	ISO3	Years	$\overline{Z}$	$\overline{\widehat{Z}}$	MAE	Success (%)	Bias $(\overline{Z}-\overline{\widehat{Z}})$	Good	Accept.	Poor
Australia	AUS	1	0.174	0.223	0.050	95.0	-0.050	✓		
Brazil	BRA	1	0.836	0.970	0.134	86.6	-0.134		+	
Bulgaria	BGR	1	0.414	0.582	0.168	83.2	-0.168		+	
Canada	CAN	3	0.461	0.528	0.067	93.3	-0.067	✓		
Chile	CHL	1	0.679	0.811	0.132	86.8	-0.132		+	
Czechia	CZE	1	0.000	0.133	0.133	86.7	-0.133		+	
Estonia	EST	1	0.103	0.396	0.292	70.8	-0.292			×
Finland	FIN	1	0.387	0.334	0.054	94.6	0.054	✓		
Germany	DEU	2	0.205	0.299	0.094	90.6	-0.094	✓		
Greece	GRC	1	0.849	0.939	0.089	91.1	-0.089	✓		
Korea	KOR	2	0.382	0.488	0.106	89.4	-0.106		+	
Mexico	MEX	2	0.835	0.925	0.090	91.0	-0.090	✓		
Netherlands	NLD	2	0.114	0.113	0.082	91.8	0.001	✓		
Poland	POL	2	0.826	0.922	0.096	90.4	-0.096	✓		
Romania	ROU	2	0.850	0.916	0.066	93.4	-0.066	✓		
Russia	RUS	2	0.373	0.624	0.250	75.0	-0.250			×
Slovak Republic	SVK	1	0.423	0.146	0.277	72.3	0.277			×
Slovenia	SVN	1	0.299	0.200	0.099	90.1	0.099	✓		
Spain	ESP	2	0.202	0.264	0.062	93.8	-0.062	✓		
Switzerland	CHE	1	0.346	0.460	0.113	88.7	-0.113		+	
Türkiye	TUR	2	0.979	0.992	0.021	97.9	-0.013	✓		
United Kingdom	GBR	1	0.312	0.406	0.094	90.6	-0.094	✓		
United States	USA	1	0.517	0.447	0.070	93.0	0.070	✓		

Note. Classification thresholds: good if MAE ≤ 0.10; acceptable if $0.10 <$ MAE ≤ 0.20; poor if MAE > 0.20.

Table S9: ***Data scope:*** *model-implied finite-difference* ΔZ *values under a +1 SD shock in each standardized modifier, evaluated at low/mid/high structural baselines. Baselines use* $(X,Y) \in \{(0.2,0.2),(0.5,0.5),(0.8,0.8)\}$ *in the structural interior band under a fixed contextual state* $(M_{\text{std}} = 0.8, T_{\text{std}} = 1.2, D_{\text{std}} = 0.6)$*, with baseline predictions* $Z \approx 0.004$*,* 0.033*, and* 0.053*.* ***Method context:*** *coefficients follow the production-pinned contract (src/data/feCoefficients.ts) and bracketed ranges are precomputed diagnostic uncertainty bands for the fixed scenario contract.* ***Interpretation boundary:*** *these are regime-dependent scenario diagnostics, not causal or policy-intervention estimates; Tier A/B inferential anchors remain refit OOS evidence, and interval width in saturated regions mainly reflects bounded-map flatness rather than transportable precision.* ***Note:*** C_{std} *is derived from WVS cultural-practice items sharing source survey with the outcome Z; the large marginal effect partially reflects this shared-source construction and should be interpreted with that overlap in mind.*

Modifier (+1 SD)	ΔZ (low baseline)	ΔZ (mid baseline)	ΔZ (high baseline)
C_{std} (cultural intensity)	0.9953 [0.1016, 0.9980]	0.9670 [0.0259, 0.9979]	0.9473 [0.0138, 0.9978]
M_{std} (media environment)	-0.0037 [-0.4142, 0.5546]	-0.0269 [-0.6299, 0.5476]	-0.0427 [-0.7050, 0.5306]
T_{std} (trauma/shock)	0.0066 [-0.0311, 0.9507]	0.0451 [-0.0997, 0.9057]	0.0687 [-0.1586, 0.8907]
D_{std} (social atomization)	0.0057 [-0.0793, 0.6853]	0.0393 [-0.2036, 0.6849]	0.0601 [-0.2034, 0.6787]
G_{std} (redistribution capacity)	-0.0006 [-0.0527, 0.0025]	-0.0045 [-0.0598, 0.0046]	-0.0070 [-0.0624, 0.0056]
E_{std} (education capacity)	-0.0044 [-0.8666, 0.0000]	-0.0328 [-0.9636, 0.0000]	-0.0525 [-0.9691, 0.0000]
S_{std} (social support)	0.0011 [-0.0927, 0.2231]	0.0078 [-0.1259, 0.2611]	0.0122 [-0.1243, 0.2589]
R_{std} (welfare capacity)	0.0305 [-0.0992, 0.8398]	0.1036 [-0.3277, 0.8192]	0.0767 [-0.4333, 0.7992]

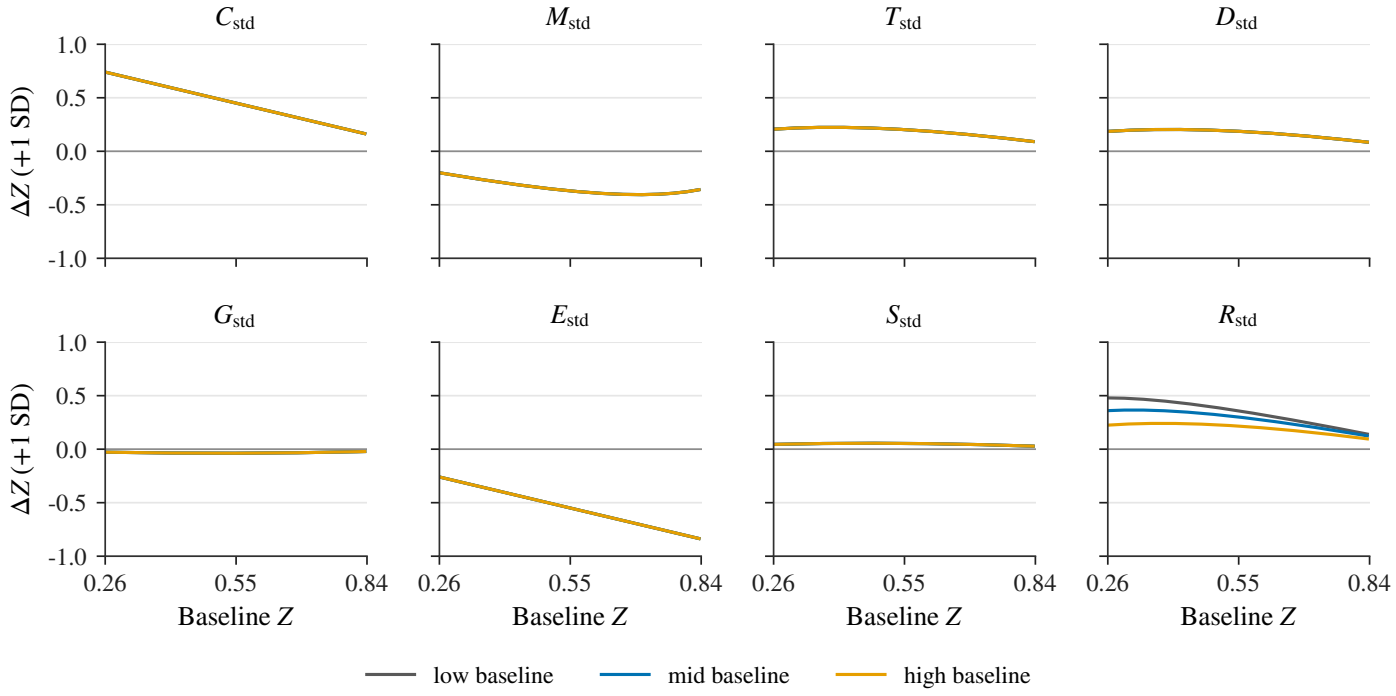
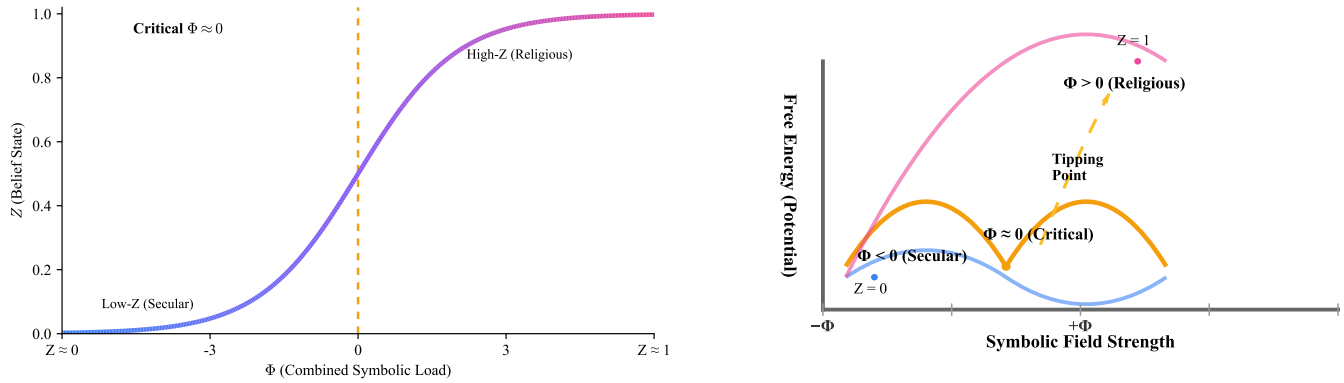


Figure S5. ***Data scope:*** *scenario-based marginal-effect curves showing ΔZ under +1 SD shocks as a function of baseline prediction Z. Curves are shown for three structural baselines in the interior band $(X,Y) \in \{(0.2,0.2),(0.5,0.5),(0.8,0.8)\}$ under the same fixed contextual state used in Table S9; the baseline-Z axis is restricted to an empirically supported interior range from sensitivity-weighted quantiles of the canonical panel.* ***Method context:*** *coefficients follow the production-pinned contract (src/data/feCoefficients.ts) and the plot exposes regime dependence under the same fixed scenario contract as the table.* ***Interpretation boundary:*** *these are model-implied regime-conditional associations, not causal response functions or Tier A/B refit OOS validation results. The regime sensitivity pattern (ΔZ peaking near $Z = 0.5$) reflects bounded-scale mapping geometry, not an independently validated behavioral threshold.*



(A) Phase-transition schematic **(B)** Energy-field intuition

Figure S6. ***Data scope:*** *documentation schematics used in the Section 3 exposition.* ***Method context:*** *Panel **(A)** shows the bounded response $Z(\Phi)$ around the transition region, and Panel **(B)** shows the potential-landscape analogy and tipping direction used to explain the mechanism.* ***Interpretation boundary:*** *these are explanatory schematics only; they are not empirical evidence, evidence-tier support, or causal identification. The transition region around $Z \approx 0.5$ reflects bounded-scale mapping geometry.*

structure. Calibration on deciles of $\widehat{Z}$ provides a coarse but robust summary of conditional central tendency and dispersion; departures from the 45° line indicate bias localized to predicted levels. Residual histograms characterize marginal fit quality, while residual-versus-predicted displays reveal potential heteroskedasticity or state-dependent misspecification. Because the evaluation is performed on the bounded support $\mathscr{I}_\varepsilon$, extreme values are clamped for numerical stability, avoiding spurious leverage from saturation regions. The first figure in this section presents the observed versus predicted scatterplot and calibration-by-decile analysis, while the second shows marginal residual distribution and residual-versus-predicted diagnostics to detect any systematic patterns in the errors. Checking for model misspecification across prediction levels is essential because non-linearity and bounded response can mask systematic bias. Examination of decile-wise calibration can identify whether certain regimes (e.g., low, medium, or high predicted levels) suffer from consistent under- or over-prediction. The width of the IQR bands also reveals whether prediction uncertainty is constant or varies with predicted level.

1. Context field (Φ)

$$\Phi = \phi_0 + \Phi_{\text{main}} + \Phi_{\text{int}} + \Phi_{\text{quad}}$$

Main: $C_{\text{std}}, M_{\text{std}}, T_{\text{std}}, D_{\text{std}}, G_{\text{std}}, E_{\text{std}}, S_{\text{std}}, R_{\text{std}}$

Interactions: XY, YR_{std}, $T_{\text{std}}R_{\text{std}}$, $C_{\text{std}}M_{\text{std}}$, $E_{\text{std}}S_{\text{std}}$, $D_{\text{std}}T_{\text{std}}$, $G_{\text{std}}R_{\text{std}}$

Quadratics: C^2_{std}, M^2_{std}, T^2_{std}, E^2_{std}

Amp (+): C, T, S, R

Sup (−): M, D, G, E

2. Inputs

Raw drivers: $X_{\text{raw}}, Y_{\text{raw}}$

Context (std): $C_{\text{std}}, M_{\text{std}}, T_{\text{std}}, D_{\text{std}}, G_{\text{std}}, E_{\text{std}}, S_{\text{std}}, R_{\text{std}}$

3. Safe band

Band map (deterministic)

$X, Y \in (\varepsilon, 1-\varepsilon)$

$X = \text{clip}(\text{minmax}(X_{\text{raw}}), \varepsilon, 1-\varepsilon)$; similarly Y

4. Stimulus

$$S = X^{\alpha} Y^{\beta} e^{\Phi}$$

$$\log S = \alpha \log X + \beta \log Y + \Phi$$

5. Saturation

$$Z = \frac{S}{S+k}$$

$S = k \dfrac{Z}{1-Z}$; k: half-saturation ($Z = 0.5$)

Figure S7. ***Data scope:*** *end-to-end computational pipeline for the implemented belief-emergence mechanism.* ***Method context:*** *the figure traces construction of Φ, deterministic preprocessing into the interior safe band, stimulus formation $S = X^{\alpha_S} Y^{\beta_S} e^{\Phi}$, and half-saturation mapping $Z = S/(S+\kappa_S)$ under the production computational contract.* ***Interpretation boundary:*** *this is a reproducibility schematic that documents how quantities are computed; it does not itself validate fit, assign an evidence tier, or support causal interpretation.*

Heteroskedasticity in the residuals would manifest as a funnel pattern in the residual-versus-predicted plot, suggesting that the model's predictive variance is not homogeneous across the support. *Reading note.* These plots should be read jointly:

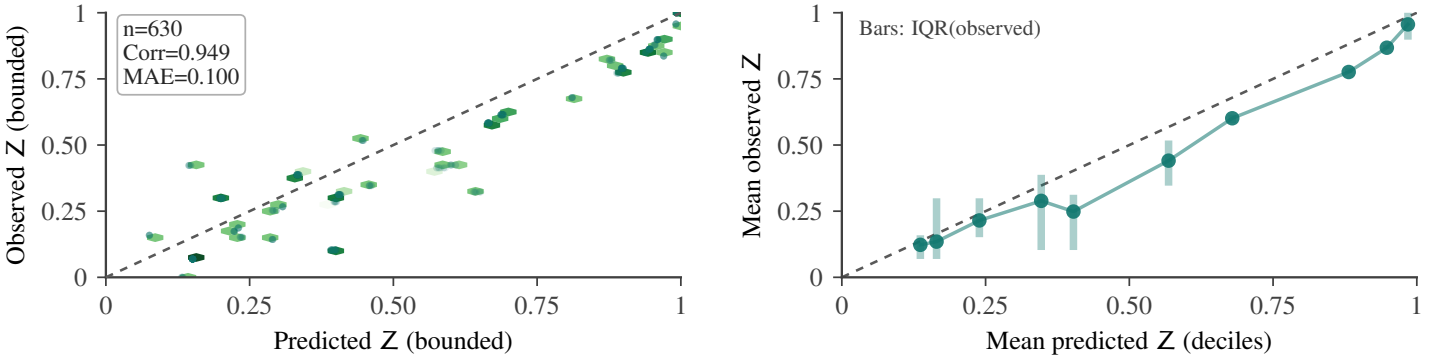


Figure S8. ***Data scope:*** *bounded-scale in-sample fit and decile-calibration diagnostics on the canonical panel. Observed outcomes Z and predictions $\widehat{Z}$ are evaluated on the same bounded scale $\mathscr{I}_\varepsilon = [\varepsilon, 1-\varepsilon]$ with $\varepsilon = 10^{-4}$, and predictions are clamped for numerical stability.* ***Method context:*** *the left panel shows observed-versus-predicted values with hexagonal support density, while the right panel reports decile-wise mean Z and within-decile* IQR(Z) *conditional on predicted level.* ***Interpretation boundary:*** *this figure provides descriptive fit context only; primary inferential weight remains Tier A/B refit OOS evidence, and the figure does not by itself support causal or policy claims.*

strong global agreement can coexist with localized calibration departures. If the calibration curve deviates systematically in the low or high deciles, that pattern should be interpreted as regime-dependent bias rather than noise, and it should be cross-checked against the residual-versus-$\widehat{Z}$ trend in the next figure. **Regime-dependent effects and correlation diagnostics**

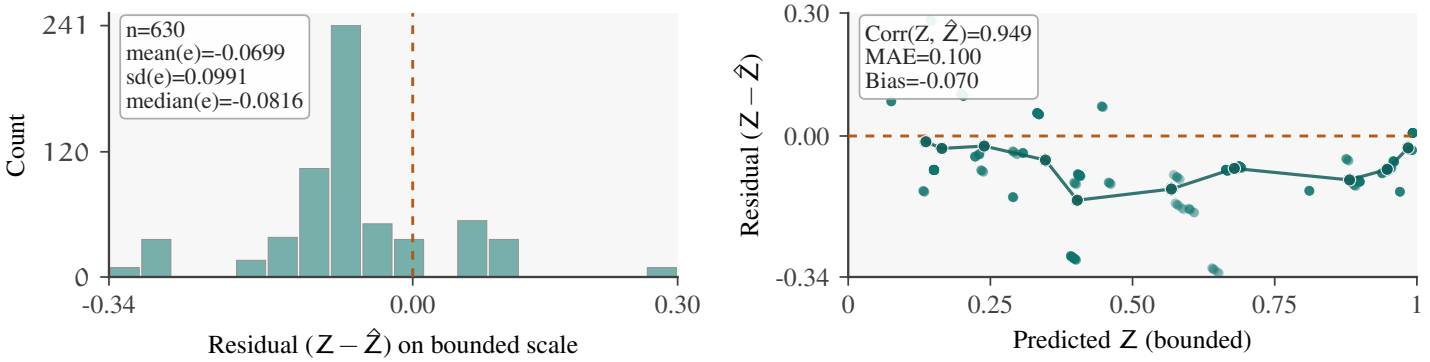


Figure S9. ***Data scope:*** *bounded-scale residual diagnostics on the canonical panel. Residuals are computed as $e_i = Z_i - \widehat{Z}_i$ using bounded observed outcomes and clamped predictions in $\mathscr{I}_\varepsilon$.* ***Method context:*** *the left panel summarizes the marginal error distribution with a histogram and zero reference line, and the right panel screens for state-dependent bias or heteroskedasticity using residuals versus predicted $\widehat{Z}$ with a decile-binned mean overlay.* ***Interpretation boundary:*** *this is a descriptive misspecification screen only; it is not primary inferential evidence and does not support causal interpretation.*

This block aggregates the manuscript's regime-aware marginal-effect visualizations and descriptive correlation diagnostics. Keeping them adjacent makes it easier to interpret effect heterogeneity in light of covariate dependence and potential collinearity structure. *Reading note.* In tables and plots that report standardized variables (e.g., C_{std}, M_{std}), the scientific interpretation is inherently relative: a +1 SD perturbation represents a move that is large enough to be comparable across contexts, but it is not necessarily large in absolute units. Accordingly, the regime panels are treated as a comparative map of *where* and *when* the model expresses sensitivity, not as an assertion that every standardized shift corresponds to an equally feasible real-world intervention. This framing is particularly important near the bounded endpoints, where the same standardized shift can produce a small ΔZ purely because the map is locally flat. Marginal effects are computed as finite differences ΔZ under +1 SD perturbations to standardized modifiers, holding a fixed contextual backdrop while varying structural baselines in the interior band. Because Z is bounded and monotone in Φ only after transformation, the *size and sign* of ΔZ naturally depend on the baseline level and local curvature of Φ. Robust variants summarize sensitivity bands, clarifying which regimes are most stable to perturbations. Correlation heatmaps provide a descriptive view of covariate dependence; they are included as context for interpreting effect sizes without conflating inference with collinearity. Figure 11 and its variants (Figures 12 main and robust) show how marginal effects vary across different regimes and parameter ranges, while Figures A2 and A3 document the overall and within-panel correlation structures to support interpretation of the empirical relationships. The importance of regime-dependent effects lies in recognizing that the bounded response is inherently non-linear. A modifier that is a strong amplifier in one region of the parameter space (e.g., where Φ is near zero and sensitivity is high) may have negligible or even inverted effects in saturation regions (where Φ is far from zero and the bounded transform is nearly flat). Robust marginal effects encompass ranges rather than point estimates, providing guidance on which regimes are most stable and informative. This multi-scale perspective prevents over-interpretation of effects in unstable regions and focuses attention on the regime where the model's predictive capability and parameter interpretability are strongest. *Reading note.* Robust marginal-effect envelopes should be interpreted as an uncertainty-aware *stability screen*: narrow bands indicate that the qualitative regime narrative is insensitive to perturbations, while wide bands flag regions where conclusions depend strongly on specification or sampling variation. In practice, the most policy-relevant regimes are those with both (i) high local sensitivity (interior band) and (ii) stable envelopes. When bands widen near saturation, this should generally be interpreted as low information rather than high disagreement: once Z is effectively flat, small numerical differences in $\widehat{Z}$ can look large relative to the remaining headroom even though substantive leverage is minimal. To avoid over-reading the tails, focus is recommended on (a) whether the sign pattern is consistent across the interior band and (b) whether the envelope remains narrow in the region where support (as indicated by the evaluation plots) is adequate. If a modifier's band is wide throughout, that modifier is best treated as an exploratory candidate rather than as a stable amplifier/suppressor narrative. Finally, note that robustness here is *conditional*: the envelopes summarize sensitivity under a fixed preprocessing and coefficient contract. They do not, by themselves, validate external transportability across countries or time periods. Accordingly, a "robust" modifier is interpreted as one whose effect is stable within the observed support and under specification perturbations, and any regime-specific narrative is treated as provisional unless it is consistent with the bounded-scale calibration and residual diagnostics. This is the same logic used throughout the tooling: explanations are considered admissible only when they align with (i) a pinned, reproducible state and (ii) evidence that the model is locally accurate in the corresponding region of $\widehat{Z}$. *Reading note.* The pooled correlation heatmap reflects both between- and within-country dependence structure, whereas the within-country heatmap removes country means and therefore emphasizes co-movement over time within each panel. This distinction matters for interpretation: strong pooled correlations can be driven by cross-sectional level differences, while within-panel correlations are more indicative of temporal coupling that could confound regime-based narratives. As a practical check, modifiers that are strongly correlated within-panel should be interpreted with additional caution in the regime-difference tables: large apparent effects may reflect shared temporal dynamics rather than separable mechanisms. **Reproducibility and additional diagnostics (moved from appendix)**

This block consolidates reproducibility notes and supporting diagnostics that are referenced from the main text as Figure/Table S#. It is included here so the anonymous main manuscript can remain within the journal's length constraints without an appendix.

Rawdata processing: source-to-column mapping. The following table provides a compact, auditable mapping from raw files under rawdata/ to the columns they feed in rawdata/panel_model_data.csv and, where relevant, to the modeled variables. It is included here to support verification of variable provenance.

Coefficient artifact key mapping. The interactive application and the deterministic evaluation scripts use a single production-pinned coefficient contract encoded in src/data/feCoefficients.ts. The UI can optionally load an external coefficient artifact at runtime (see the schema in src/utils/validation/coefficientsSchema.ts). Both sources use the same internal coefficient keys (e.g., `c1`, `c_cm`) while coefficients are reported using mathematical symbols in Eq. (4). The table

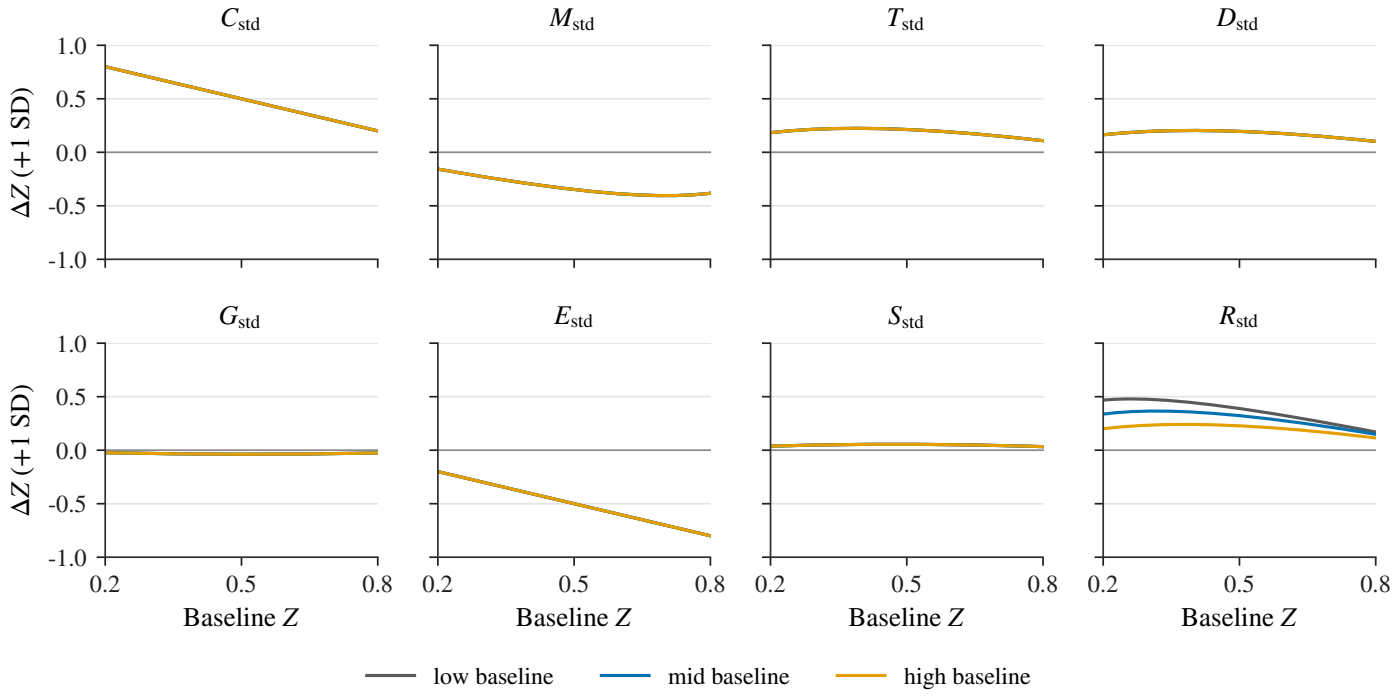


Figure S10. ***Data scope:*** *robustness rendering of the same marginal-effect curves as Fig. S5, shown on a fixed normalized baseline-Z range* $[0.2, 0.8]$. ***Method context:*** *the fixed axis improves readability and visual comparability across modifiers without redefining the empirically supported baseline-Z band used in the main figure.* ***Interpretation boundary:*** *this is a visualization-only robustness view, not a separate inferential family or causal evidence.*

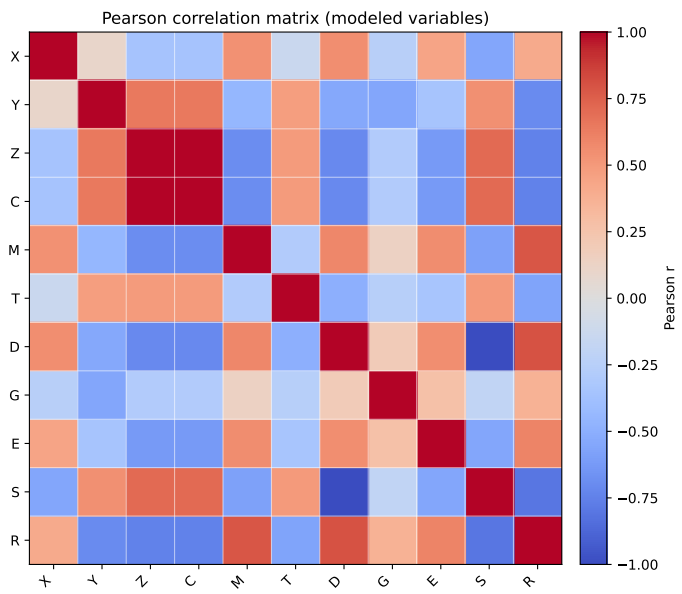


Figure S11. ***Data scope:*** *Pearson correlation matrix for the modeled variables on complete-case rows using the same required-column filtering rule as the estimation build step.* ***Method context:*** *the matrix provides a descriptive diagnostic of linear association and potential collinearity in the modeled variable block (generated by paper/scripts/make_fig_correlation_heatmap.py).* ***Interpretation boundary:*** *it is not a validation claim, evidence-tier result, or causal statement.*

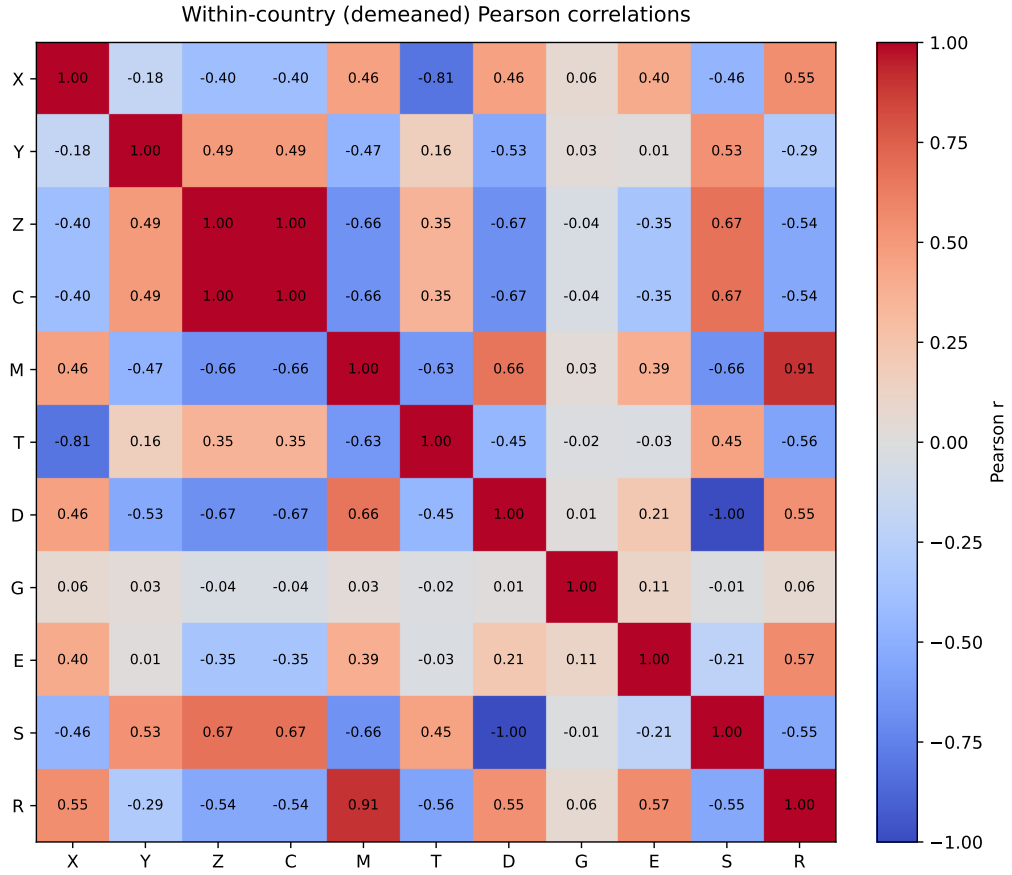


Figure S12. ***Data scope:*** *within-country (demeaned) Pearson correlation matrix for the modeled variable block* $X, Y, Z, C, M, T, D, G, E, S, R$ *on complete-case rows.* ***Method context:*** *demeaning removes country fixed effects to emphasize within-panel dependence structure.* ***Interpretation boundary:*** *this is descriptive context for reading regime narratives and possible confounding, not validation or causal evidence.*

below provides the key-to-symbol mapping.

Additional supporting tables. The following tables document slice-wise evaluation stability and descriptive collinearity diagnostics under the fixed preprocessing/coefficients contract. These appendixed fixed-model tables are descriptive diagnostics only and are not part of the primary refit OOS inferential evidence family used for Tier A/B claims.

Data scope: *in-sample bounded-scale evaluation metrics by country slice with no refit.* ***Method context:*** *metrics are computed per country on rows with finite observed and predicted Z under the fixed preprocessing and coefficient contract; em-dashes indicate undefined metrics or sample-size rule failures.* ***Interpretation boundary:*** *this is a descriptive heterogeneity screen only, not Tier A/B refit OOS evidence, transportability, or causal identification.*

Country	nUsed / nTotal	R^2	r	MAE	RMSE	Bias	Residual P10	Residual P50	Residual P90
ALL	630 / 630	0.338	0.949	0.6711	0.7873	0.3954	−0.4867	0.4393	1.2518
NLD	90 / 90	-209.260	−1.000	1.2158	1.2205	1.2158	1.2153	1.2512	1.2518
CAN	81 / 81	0.300	0.999	0.2124	0.3420	0.2124	0.0794	0.1547	0.9634
POL	72 / 72	-9.125	0.999	0.3486	0.3598	−0.3486	−0.4388	−0.3486	−0.2580
TUR	45 / 45	-224.847	0.957	0.7962	0.7979	−0.7962	−0.8227	−0.8225	−0.6909
EST	36 / 36	-9.798×10^{30}	0.000	1.3901	1.3901	1.3901	1.3860	1.3899	1.3943
FIN	36 / 36	-1.105×10^{32}	0.000	0.4376	0.4376	0.4376	0.4359	0.4368	0.4396
GBR	36 / 36	-1.807×10^{32}	0.000	0.7461	0.7461	0.7461	0.7441	0.7461	0.7480
SVN	36 / 36	-1.100×10^{32}	0.000	0.5821	0.5822	0.5821	0.5804	0.5815	0.5846
MEX	27 / 27	—	—	—	—	—	—	—	—
DEU	18 / 18	—	—	—	—	—	—	—	—
ESP	18 / 18	—	—	—	—	—	—	—	—
KOR	18 / 18	—	—	—	—	—	—	—	—
ROU	18 / 18	—	—	—	—	—	—	—	—
RUS	18 / 18	—	—	—	—	—	—	—	—
AUS	9 / 9	—	—	—	—	—	—	—	—
BGR	9 / 9	—	—	—	—	—	—	—	—
BRA	9 / 9	—	—	—	—	—	—	—	—
CHE	9 / 9	—	—	—	—	—	—	—	—

Country	nUsed / nTotal	R^2	r	MAE	RMSE	Bias	Residual P10	Residual P50	Residual P90
CHL	9 / 9	—	—	—	—	—	—	—	—
CZE	9 / 9	—	—	—	—	—	—	—	—
GRC	9 / 9	—	—	—	—	—	—	—	—
SVK	9 / 9	—	—	—	—	—	—	—	—
USA	9 / 9	—	—	—	—	—	—	—	—

Computed per country on rows with finite observed and predicted Z; no refit (fixed coefficients unless an estimated set is active). Countries with fewer than 30 valid observations may display "—" due to insufficient sample size. nUsed counts rows with finite observed and predicted Z. Values are shown as "—" when metrics are undefined or when the minimum per-country sample-size rule applies (e.g., nUsed < 30). **Visual analytics tooling (dashboard, pinned inspection, export contract)**

The following UI and workflow figures document the inspection surface used in the study: how regimes are explored, how local sensitivity is pinned and compared, and how deterministic PDF exports preserve a reproducible mapping from UI state to evidence bundles. These figures provide transparency into the interactive analysis process and demonstrate how the computational model is operationalized for scientific inspection. Interactive visual analytics are central to modern quantitative social science, enabling researchers to rapidly explore parameter space, test hypotheses, and communicate findings with supporting evidence. The dashboard surfaces the decomposition of Φ and derived quantities, enabling pinned inspection of local states and side-by-side sensitivity comparison at identical parameter snapshots. A schema-versioned export workflow serializes the interactive state vector and renderer configuration, guaranteeing that visual artifacts are exactly reproducible. This audit trail bridges interactive exploration and paper-consistent, deterministic exports. The dashboard visualization (Figure in this section) shows the main inspection interface, while the pinned inspection figure demonstrates how two states are compared side-by-side, and the export workflow figure documents the serialization process that ensures reproducibility of all generated figures. Maintaining a complete audit trail enables other researchers to replicate, verify, or extend the analyses from first principles.

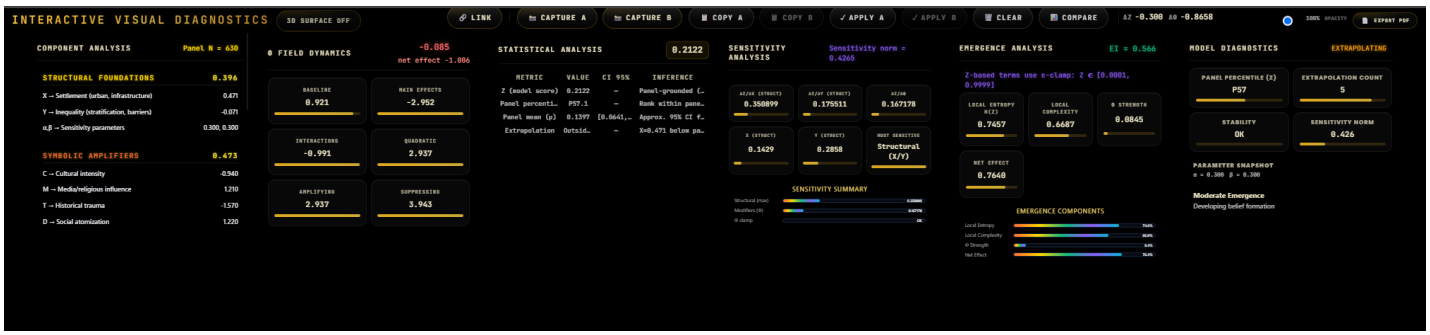


Figure S13. ***Data scope:*** *overview screenshot of the interactive visual analytics dashboard.* ***Method context:*** *the interface documents bounded-transform surface exploration, pinned/slice-based inspection, and deterministic export under the same numerical safety contract used in production.* ***Interpretation boundary:*** *this is a provenance screenshot only; it is not inferential evidence and does not substitute for manuscript evaluation figures.*

Reading note. Pinned inspection and export are treated as linked components of the evidence workflow: pinning defines the exact coordinates in (Φ, s) (and UI state vector) that a claim refers to, and exporting makes that state portable. This reduces the risk of "unrecorded" interactive choices when translating exploratory visual analytics into publication-grade figures. In methodological terms, the pair enforces a provenance constraint: every visual claim is anchored to a concrete state vector that can be replayed and verified by a third party.

Export contract and reproducibility. The export workflow serializes the complete interactive state vector (preprocessing bounds, coefficient bundle, pinned inspection coordinates, and renderer options) under a schema-versioned contract. Each PDF export is accompanied by a deterministic snapshot and a metadata header (state hash and schema version) so that regenerated artifacts are bitwise-identical to the original evidence. This design separates (i) the statistical model and bounded map $Z = \sigma(\Phi + s)$ from (ii) the inspection UI, while making the bridge between them auditable. As a result, any figure shown in this supplementary file can be reproduced from the same snapshot without relying on hidden UI conditions. **Supplementary UI screenshots (additional evidence snapshots)**

These screenshots provide supporting context for the interactive analysis environment. They are included as an audit trail of UI state and diagnostic panels; *no resizing is applied* beyond the manuscript's standard `width=\linewidth` convention. Each screenshot captures a distinct operational state of the interactive system, documenting how the model parameters,

Table S10: ***Data scope:*** *source → column provenance map for the integrated panel (rawdata/panel_model_data.csv).* ***Method context:*** *Model indicates which of $X, Y, Z, C, M, T, D, G, E, S, R$ each source contributes to directly or via deterministic transforms, and panel-wide coverage is summarized in rawdata/panel_model_data.json.* ***Interpretation boundary:*** *this is provenance documentation only; it is not inferential evidence or a quality claim by itself.*

Source (rawdata/)	Raw file(s) (as committed)	Contributed columns in panel_model_data.csv	Model variables (X,Y,Z,C,M,T,D,G,E,S,R)
World Values Survey (WVS) constructs	rawdata/wvs_combined.csv (built from WVS/EVS extracts, using config/wvs_items.yaml)	`Z_religion_importance`, `Z_violence_justifiable`, `Z_belief_outcome`, `C_religiosity_index`, `C_cultural_intensity`, `D_social_trust`, `D_social_atomization`, `S_support_index`, `S_social_support`, (plus auxiliary WVS fields)	Z, C, D, S
World Bank WDI (macro)	rawdata/worldbank_data_new.csv and WDI bulk exports (rawdata/WDICSV.csv, metadata files)	`urban_pop_share`, `gdp_ppp`	X (from `urban_pop_share`)
World Bank Internet usage	rawdata/worldbank_internet_usage.csv and API export rawdata/API_IT.NET.USER.ZS_DS2_en_csv_v2_280824.csv	`internet_users_pct`	M (via deterministic standardization to `M_media_environment`)
OECD inequality (Gini levels)	rawdata/oecd_gini.csv (derived from OECD WISE/IDD extracts)	`gini_index`, `gini_disposable`	Y (via deterministic transform to `Y_gini`)
OECD redistribution scenarios	rawdata/oecd_redistribution.csv, rawdata/OECD_Redistribution_derived.csv	`market_gini`, `disposable_gini`, `redistribution_effect`, `redistribution_pct`	G (via deterministic transform to G_redistribution_capacity); scenario replication keys
OECD SOCX (social spending)	rawdata/oecd_socx.csv (plus rawdata/SOCX2025-All-nat-cur.xlsx as upstream export)	`socx_public_social_spending_gdp`	R (via deterministic transform to `R_welfare_capacity`)
UNDP Human Development (HDR)	rawdata/undp_hdi.csv (from HDR exports)	`hdi`	R (via deterministic transform to `R_welfare_capacity`)
Education capacity proxy	(as encoded in the integrated panel)	`sec_enroll_gross`	E (via deterministic transform to E_education_capacity)
UCDP conflict deaths	rawdata/ucdp_conflict_deaths.csv (plus rawdata/GEDEvent_v25_1.csv as upstream export)	`total_battle_deaths`	T (via deterministic transform to `T_trauma_shock`)
EM-DAT disasters	rawdata/emdat_disasters.csv (plus public EM-DAT request exports)	`total_affected`, `total_deaths`	T (via deterministic transform to `T_trauma_shock`)

Table S11: ***Data scope:*** *mapping between internal coefficient keys and the terms used in Eq. (4).* ***Method context:*** *the table connects the production-pinned coefficient contract in src/data/feCoefficients.ts to manuscript notation so UI, scripts, and paper outputs can be audited against the same artifact names.* ***Interpretation boundary:*** *this is a nomenclature bridge only, not empirical evidence or validation.*

Artifact key	Eq. (4)	Term	Artifact key	Eq. (4)	Term
`c1`	c_C	C_{std}	`c_c2`	c_{c2}	C^2_{std}
`c2`	c_M	M_{std}	`c_m2`	c_{m2}	M^2_{std}
`c3`	c_T	T_{std}	`c_t2`	c_{t2}	T^2_{std}
`c4`	c_D	D_{std}	`c_e2`	c_{e2}	E^2_{std}
`c5`	c_G	G_{std}	`alpha`	α	Elasticity on $\log X$
`c6`	c_E	E_{std}	`beta`	β	Elasticity on $\log Y$
`c7`	c_S	S_{std}	`k`	k	Half-saturation constant
`c8`	c_R	R_{std}	`phi0`	ϕ_0	Baseline offset
`c_xy`	c_{xy}	XY			
`c_yr`	c_{yr}	$Y R_{\text{std}}$			
`c_tr`	c_{tr}	$T_{\text{std}} R_{\text{std}}$			
`c_cm`	c_{cm}	$C_{\text{std}} M_{\text{std}}$			
`c_es`	c_{es}	$E_{\text{std}} S_{\text{std}}$			
`c_dt`	c_{dt}	$D_{\text{std}} T_{\text{std}}$			
`c_gr`	c_{gr}	$G_{\text{std}} R_{\text{std}}$			

Table S12: ***Data scope:*** *fixed-coefficient bounded-scale evaluation metrics by temporal slice.* ***Method context:*** *rows compare all years and three time blocks under the same preprocessing and coefficient contract, with nUsed counting rows with finite observed and predicted Z.* ***Interpretation boundary:*** *this table is a descriptive slice diagnostic only; it is not refit OOS evidence or causal support.*

Period	Years	nUsed / nTotal	R^2	Corr	MAE	RMSE	Bias	Residual P10	Residual P50	Residual P90
all years	—	630 / 630	0.846	0.949	0.1005	0.1212	0.0699	−0.0568	0.0816	0.1491
≤ 2007	≤ 2007	261 / 261	0.808	0.962	0.0855	0.0890	0.0378	−0.0964	0.0808	0.1110
2008–2013	>2007 to ≤ 2013	252 / 252	0.857	0.972	0.1148	0.1469	0.1127	−0.0073	0.0818	0.2922
>2013	>2013	117 / 117	0.836	0.940	0.1030	0.1223	0.0495	−0.0712	0.0557	0.1341

nUsed counts rows with finite observed and predicted Z.

Table S13: ***Data scope:*** *variance-inflation-factor diagnostics for the model input block on complete-case rows.* ***Method context:*** *VIFs are computed on* $\log X$, $\log Y$, *and z-scored* C, M, T, D, G, E, S, R *to summarize collinearity in the estimation design matrix.* ***Interpretation boundary:*** *this is stability context only; high VIF does not by itself invalidate fitted results or establish causal structure.*

Variable	VIF
D	∞
S	∞
R	7.28
M	4.74
C	3.53
$\log X$	2.63
$\log Y$	2.17
E	1.99
T	1.72
G	1.61

Notes: computed on complete-case rows (N=630); inputs are $\log X$, $\log Y$, and C, M, T, D, G, E, S, R (z-scored).

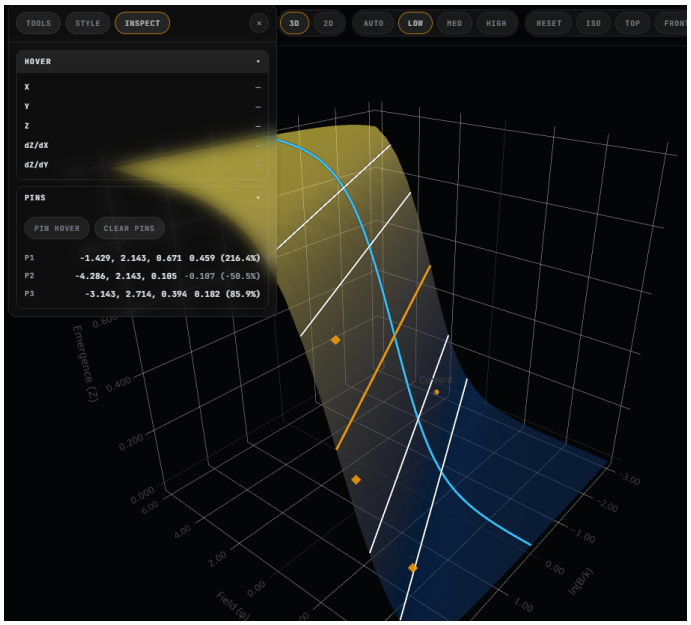


Figure S14. ***Data scope:*** *screenshot of local pinned-point inspection on the 3D response surface.* ***Method context:*** *the UI exposes pinned coordinates, local predicted levels, gradient components, and regime labels for a fixed state under the deterministic contract, allowing pointwise comparison of local sensitivity on the bounded surface.* ***Interpretation boundary:*** *this is provenance and inspection-interface documentation only; it aids local explanation but is not Tier A/B evidence or causal identification.*

decomposition, and derived diagnostics are presented to analysts during the exploration process. Screenshots are included to document representative parameter states and diagnostic panels, complementing the main evaluation by showing how the mechanism-level decomposition and checks appear within the analysis surface. They do not alter the computational contract; rather, they provide additional, verifiable context for the interpretation pipeline and export protocol. The first screenshots show alternative UI states, followed by explanatory paragraphs describing key inspection techniques (geometry reading, curvature visualization, preset management), and ending with screenshots of specialized diagnostic panels (belief equation, core variables, amplifiers, suppressors, mathematical analysis, system dynamics, and AI interpretation layer). Transparency in analytical methods is a cornerstone of reproducible science. Documenting the interactive UI state, parameter values, and diagnostic outputs enables readers to understand not just the final conclusions but the entire process by which results were obtained. The screenshots serve as a visual record of the analytical workflow, from initial parameter setting through iterative sensitivity exploration to final evidence generation. This comprehensive documentation supports scientific scrutiny

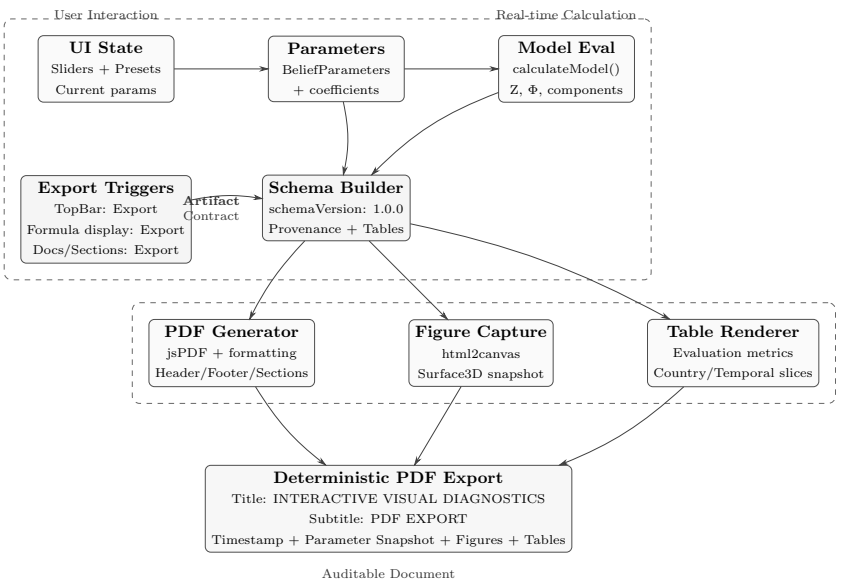


Figure S15. ***Data scope:*** *schematic of the PDF export workflow from UI state to compiled artifact.* ***Method context:*** *the export routine serializes a schema-versioned state snapshot that parameterizes downstream renderers while reusing fixed estimation artifacts, preserving the mapping between interactive state and report content.* ***Interpretation boundary:*** *this is a reproducibility-contract figure only; it is not model validation or inferential evidence.*

and allows future researchers to build upon the methodology with full confidence in the computational integrity of the reported results. The following screenshots have been selected to represent key operational states of the analysis UI. Each screenshot is accompanied by technical captions describing the analytical purpose and the displayed diagnostics. *Reading note.* The primary interpretive role of these screenshots is provenance rather than persuasion: they document what an analyst could see at the time a state was pinned and exported. Accordingly, the captions focus on the bounded-scale contract, the decomposition of Φ, and the mechanisms used to prevent misreading saturation regions as substantive regime shifts. In combination with the deterministic export workflow, the screenshots also clarify where a given narrative came from: the key objects are (i) the state vector, (ii) the term-by-term Φ contributions, and (iii) the bounded-scale sensitivity at that state. This is intended to make reviewer verification feasible even without re-running interactive exploration from scratch. Because screenshots are "thick" evidence (they embed multiple panels at once), short interpretive guidance is included between blocks to indicate which parts of the UI correspond to which scientific checks (sensitivity, decomposition, and calibration). *Reading the geometry view.* The geometry surface displays the bounded response Z across the (X,Y) stimulus plane under a fixed contextual backdrop, with slices and overlays indicating curvature and sensitivity bands. Because Z is the monotone transform of Φ, apparent slope and saturation reflect local field intensity rather than raw additive scale. Pinned inspection compares two states with identical preprocessing and coefficients, isolating the effect of baseline shifts; this makes regime-dependent differences interpretable without conflating them with scaling artefacts. *Reading note.* When comparing screenshots (3) and (4), the relevant scientific question is not whether the surface "looks" different, but whether the change corresponds to a controlled variation in the state vector. The pinned-inspection mechanism is intended to enforce this control, so that qualitative differences can be traced to specific term-level contributions rather than implicit rescaling or navigation artifacts. In practice, this means that any narrative comparison should be accompanied by an explicit statement of what was held fixed (preprocessing bounds, coefficients, and baseline (X,Y)) and what was changed (a specific modifier or preset), mirroring the finite-difference logic used in the regime-effect tables. When this control is respected, screenshots can be treated as a compact experiment log for interpretability: they capture not only outcomes, but also the intermediate quantities that justify a mechanistic reading. In practical review workflows, the most efficient use of these panels is to verify three items in sequence: (i) the state identity (preset name or pinned coordinates), (ii) the decomposition (which terms account for the net Φ shift), and (iii) the sensitivity (whether the local slope $\partial Z/\partial\Phi$ supports a meaningful change on the bounded scale). If (i) and (ii) hold but (iii) indicates saturation, then changes in the surface may be visually salient but scientifically low-impact; conversely, when sensitivity is high, small compositional shifts can legitimately justify a large ΔZ narrative. This discipline aligns the screenshot evidence with the finite-difference tables and prevents qualitative over-interpretation driven by display curvature alone. *Operational guideline.* A useful way to treat each screenshot pair is as a two-condition sensitivity test: identify the single parameter or toggle that differs across states, and then ask whether the predicted change is aligned with the sign and regime pattern implied by the marginal-effect curves. If a screenshot suggests a large qualitative shift but the corresponding finite-difference curve is flat in that regime, the most likely explanation is that the state lies on a low-sensitivity shoulder (bounded saturation) or that the two states are not truly comparable (hidden changes in preprocessing bounds or coefficients). Conversely, when both the screenshot comparison and the curve agree (same sign, same high-sensitivity region), the interpretation is strengthened because two independent views of the same

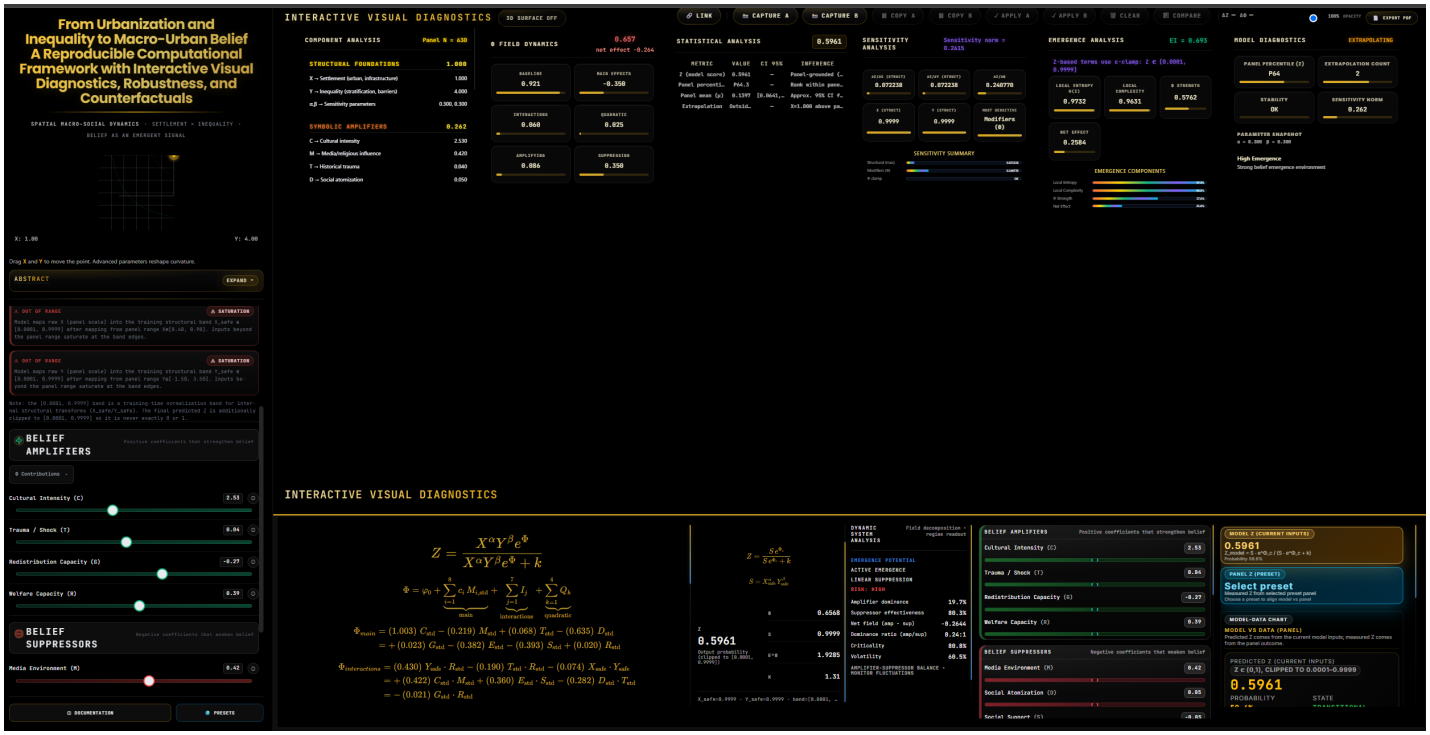


Figure S16. ***Data scope:*** *supplementary UI screenshot showing one alternative interactive state snapshot.* ***Method context:*** *the view records a concrete slider/configuration state used to inspect regime dependence of the bounded response Z under the fixed preprocessing and coefficient contracts.* ***Interpretation boundary:*** *this is a provenance screenshot only, included for auditability rather than inferential evidence or causal support.*

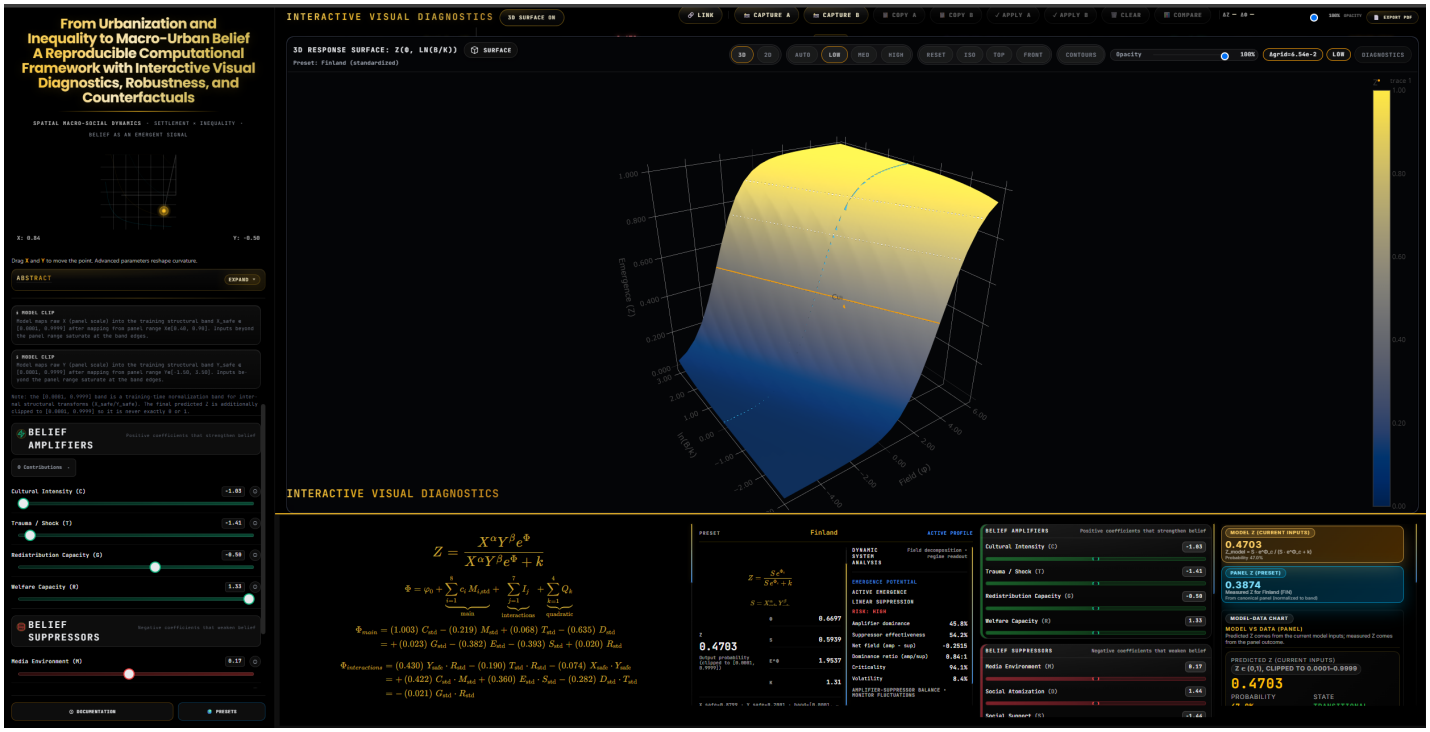


Figure S17. ***Data scope:*** *supplementary UI screenshot showing a second interactive state snapshot.* ***Method context:*** *it documents a distinct configuration of controls, surface view, and derived diagnostics under the same preprocessing and coefficient contracts.* ***Interpretation boundary:*** *this is provenance documentation only; it enables replay of the displayed state but does not provide inferential evidence.*

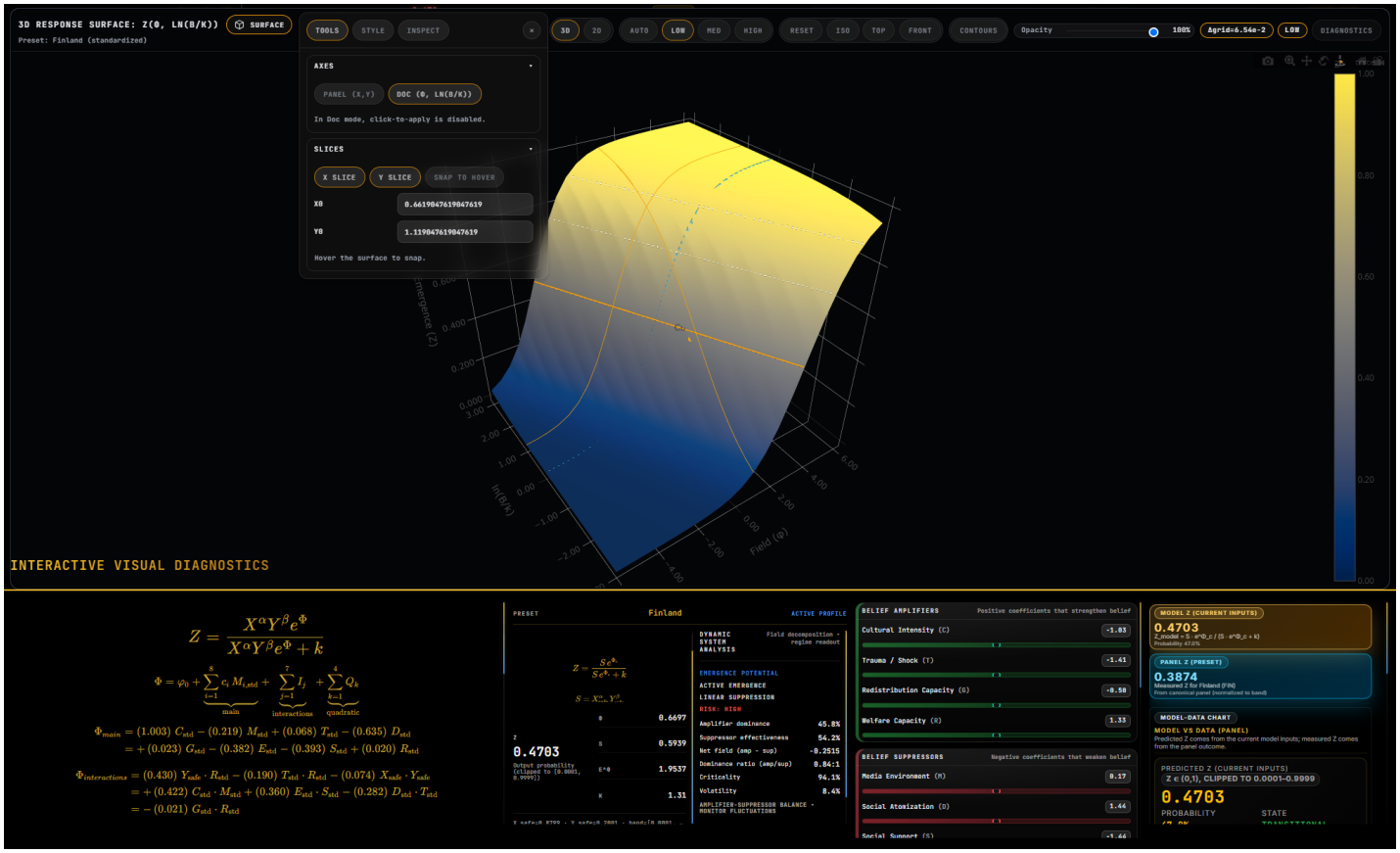
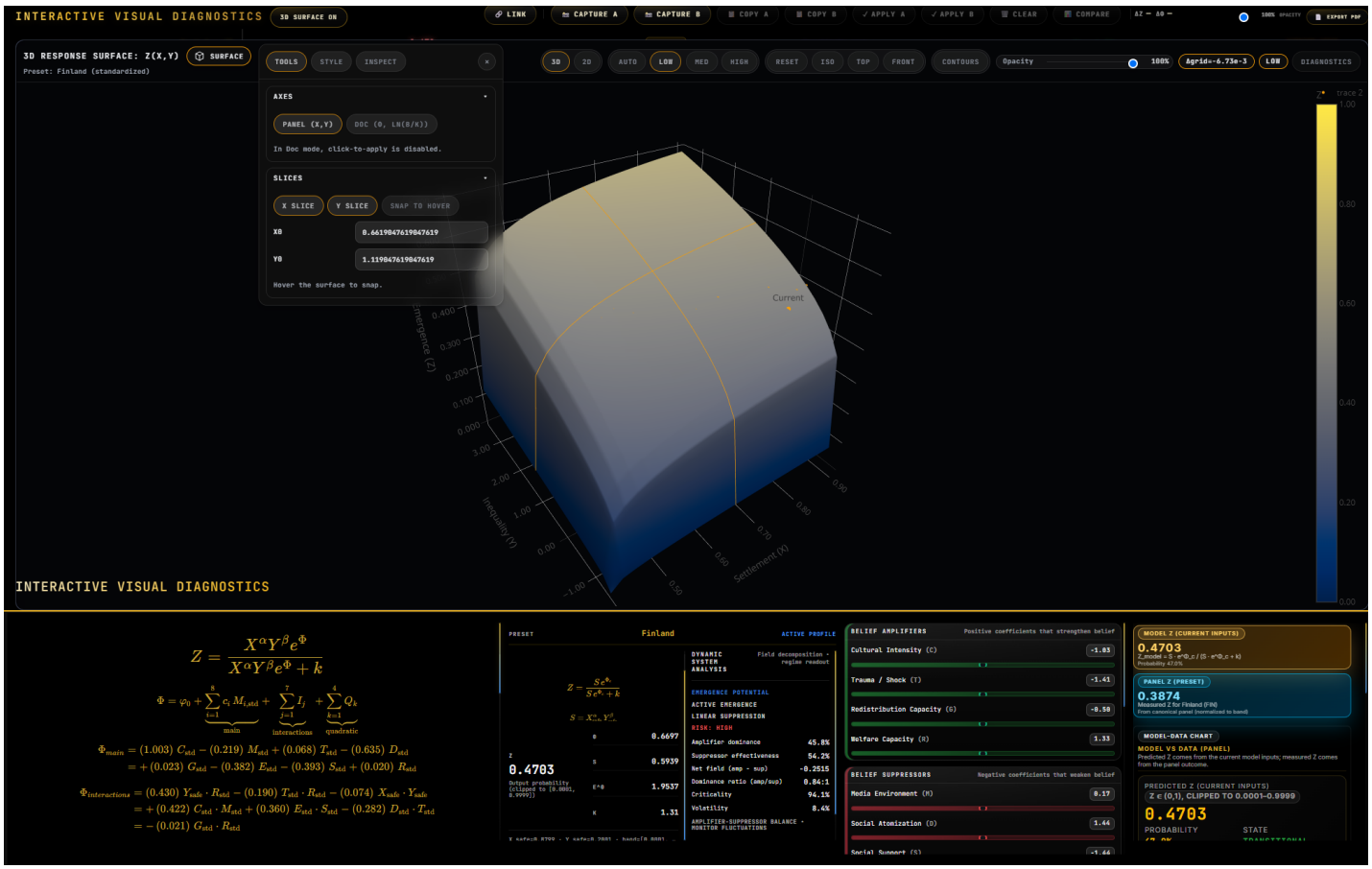


Figure S18. ***Data scope:*** *supplementary UI screenshot showing a third interactive state snapshot.* ***Method context:*** *the capture documents how the UI presents model geometry and derived quantities for a specific deterministic parameter state.* ***Interpretation boundary:*** *this is an auditability record only, not a standalone inferential or causal figure.*

deterministic contract are consistent. In practice, the most compact way to make this check auditable is to record the preset name or state hash in the export metadata and confirm it is identical across the two conditions. *Curvature and sensitivity bands.* The UI overlay on the geometry surface shows predicted regions of high and low sensitivity, rendered as confidence or uncertainty bands around the central trajectory. These bands reflect the local slope and stability of the bounded transform near the current prediction level, making it visually clear where marginal effects are steep versus flat. Bright saturated regions indicate that small baseline perturbations produce large bounded responses, whereas pale or white regions indicate saturation or near-boundary behavior where the bounded map is less responsive. *Why this matters for interpretation.* Because the mapping $Z = \sigma(\Phi + s)$ is bounded and non-linear, *two* states can have very different decompositions while yielding similar $\widehat{Z}$ if both lie in a locally flat region. For mechanistic narratives, the appropriate evidentiary standard is therefore twofold: (i) the decomposition must change in the hypothesized direction and (ii) the local sensitivity must be high enough that the bounded outcome can plausibly move. The sensitivity overlays provide this second check directly at the point of inspection, reducing reliance on intuition about what a "large" visual change means. In the analysis workflow, these overlays serve as a guardrail: they prompt the analyst to prefer explanations rooted in the interior band (where $\partial Z/\partial\Phi$ is non-negligible) and to treat near-boundary visual variation as potentially cosmetic. This is especially important when communicating amplifier/suppressor narratives, since term contributions to Φ can remain large even when the bounded response is locally insensitive. *Interpretive guardrail.* When a surface appears visually dramatic, the question to ask is whether the state lies near the transition ridge (high sensitivity) or on a saturation shoulder (low sensitivity). In the former case, moderate changes in Φ can legitimately imply a large bounded-scale movement; in the latter, similar changes should be interpreted as shifts in decomposition with limited outcome leverage. The preset workflow below is designed to make this check routine by keeping the baseline state fixed and making any perturbation explicit. *Reading note.* The preset mechanism below operationalizes this guardrail by packaging controlled starting points: the same named state can be reloaded, pinned, and exported so that any downstream figure is tied to an identical parameterization. This makes regime comparisons reproducible and limits post-hoc reinterpretation driven by accidental navigation to a different sensitivity region. As a consequence, disagreements between two narrative readings can be adjudicated by reloading the same preset and checking whether (i) the decomposition changes as claimed and (ii) the local sensitivity supports the claimed bounded-scale impact. *Preset mechanism and reproducibility.* Presets encode fixed parameter bundles as named snapshots within the interactive state vector. When a preset is activated, all



$$Z = \frac{X^{\alpha} Y^{\beta} e^{\Phi}}{X^{\alpha} Y^{\beta} e^{\Phi} + k}$$

$$\Phi = \varphi_0 + \underbrace{\sum_{i=1}^{8} c_i M_{i,\text{std}}}_{\text{main}} + \underbrace{\sum_{j=1}^{7} I_j}_{\text{interactions}} + \underbrace{\sum_{k=1}^{4} Q_k}_{\text{quadratic}}$$

$$\begin{aligned}\Phi_{main} &= (1.003)\, C_{\text{std}} - (0.219)\, M_{\text{std}} + (0.068)\, T_{\text{std}} - (0.635)\, D_{\text{std}} \\ &= +(0.023)\, G_{\text{std}} - (0.382)\, E_{\text{std}} - (0.393)\, S_{\text{std}} + (0.020)\, R_{\text{std}}\end{aligned}$$

$$\begin{aligned}\Phi_{interactions} &= (0.430)\, Y_{\text{safe}} \cdot R_{\text{std}} - (0.190)\, T_{\text{std}} \cdot R_{\text{std}} - (0.074)\, X_{\text{safe}} \cdot Y_{\text{safe}} \\ &= +(0.422)\, C_{\text{std}} \cdot M_{\text{std}} + (0.360)\, E_{\text{std}} \cdot S_{\text{std}} - (0.282)\, D_{\text{std}} \cdot T_{\text{std}} \\ &= -(0.021)\, G_{\text{std}} \cdot R_{\text{std}}\end{aligned}$$

Figure S19. ***Data scope:*** *supplementary UI screenshot showing an additional surface-instrument inspection state.* ***Method context:*** *the capture documents UI rendering under the bounded-transform safety logic and fixed coefficient artifacts during interactive exploration.* ***Interpretation boundary:*** *this is provenance documentation only; it does not introduce new inferential evidence.*



Figure S20. ***Data scope:*** *supplementary preset-configuration views for reproducible UI state vectors.* ***Method context:*** *the views document how named preset bundles synchronize sliders and toggles before comparing Φ decomposition and the resulting bounded response surface Z.* ***Interpretation boundary:*** *this is a provenance/control-mechanism figure only, not inferential evidence.*

sliders, toggles, and diagnostic view settings are synchronized to the preset's stored configuration, ensuring that comparisons across two or more states reflect only the intended variation. The preset labels (e.g., "baseline regime," "high urbanization scenario") are user-defined and serve as mnemonics for interpretable scenario labels; the underlying numerical values are deterministically exported alongside the figure, guaranteeing external reproducibility. *Reading note.* The amplifier and



Figure S21. ***Data scope:*** *screenshot of the belief-equation display panel.* ***Method context:*** *the UI renders the computational form of the belief field Φ and the bounded transform as an inspectable object linking interactive parameters to additive components and the downstream response Z.* ***Interpretation boundary:*** *this is interpretability-layer documentation only, not independent evidence or causal support.*

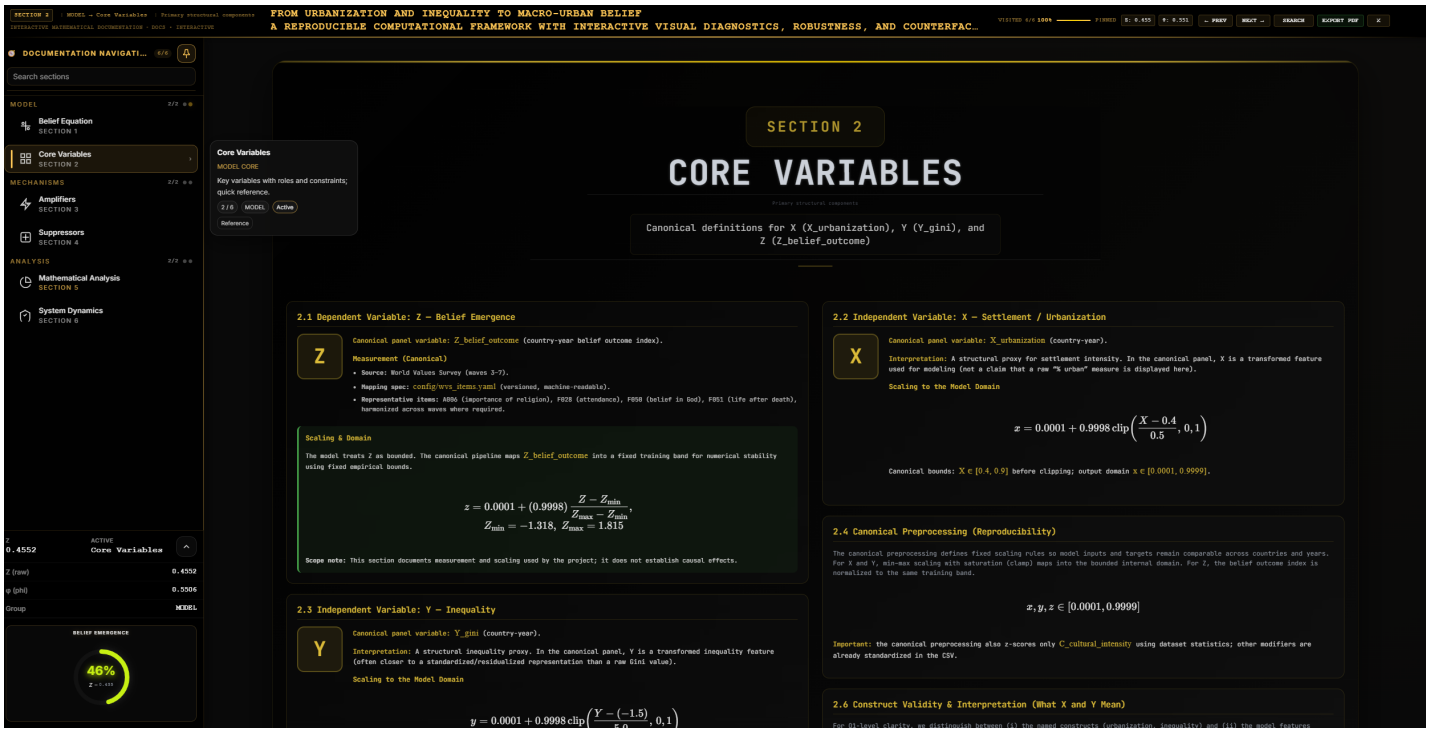


Figure S22. ***Data scope:*** *screenshot of the core-variable control panel.* ***Method context:*** *the panel documents the standardized controls exposed for interactive exploration of the (X, Y) stimulus terms under fixed preprocessing bounds.* ***Interpretation boundary:*** *this is UI provenance only; it is not inferential evidence.*

suppressor panels should be interpreted as paired views of the same additive decomposition: a term can move between

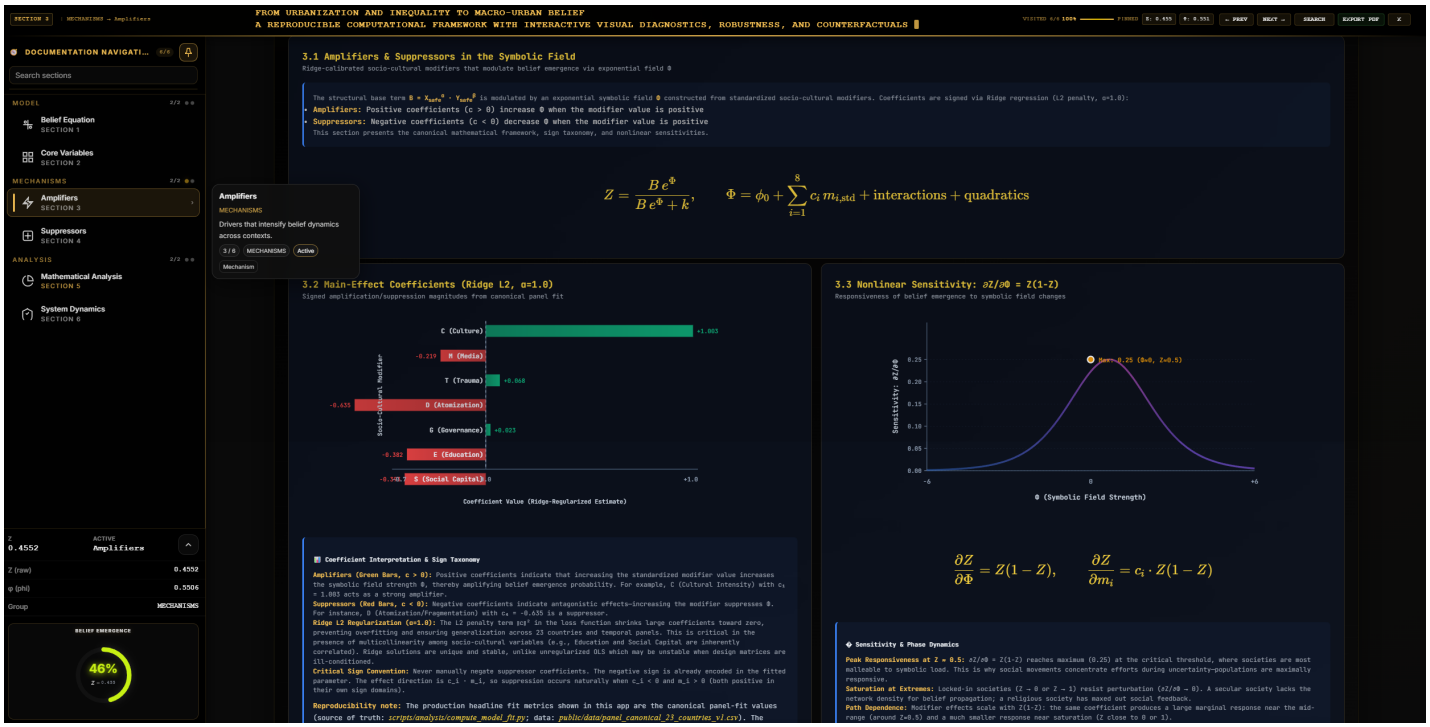


Figure S23. ***Data scope:*** *screenshot of the belief-amplifier panel.* ***Method context:*** *the UI decomposes Φ into signed additive components and surfaces positive state-specific contributions under the current deterministic state vector.* ***Interpretation boundary:*** *this is a local interpretability aid only; it is not global sign evidence, Tier A/B validation, or causal proof.*

panels across regimes as the state changes, even when its global coefficient sign is fixed. This is expected in interaction-rich specifications, and it is precisely why state-anchored, deterministic inspection is emphasized rather than universal sign narratives. *Reading note.* This diagnostics panel is intended as a convenience aggregation: it helps verify that a pinned state

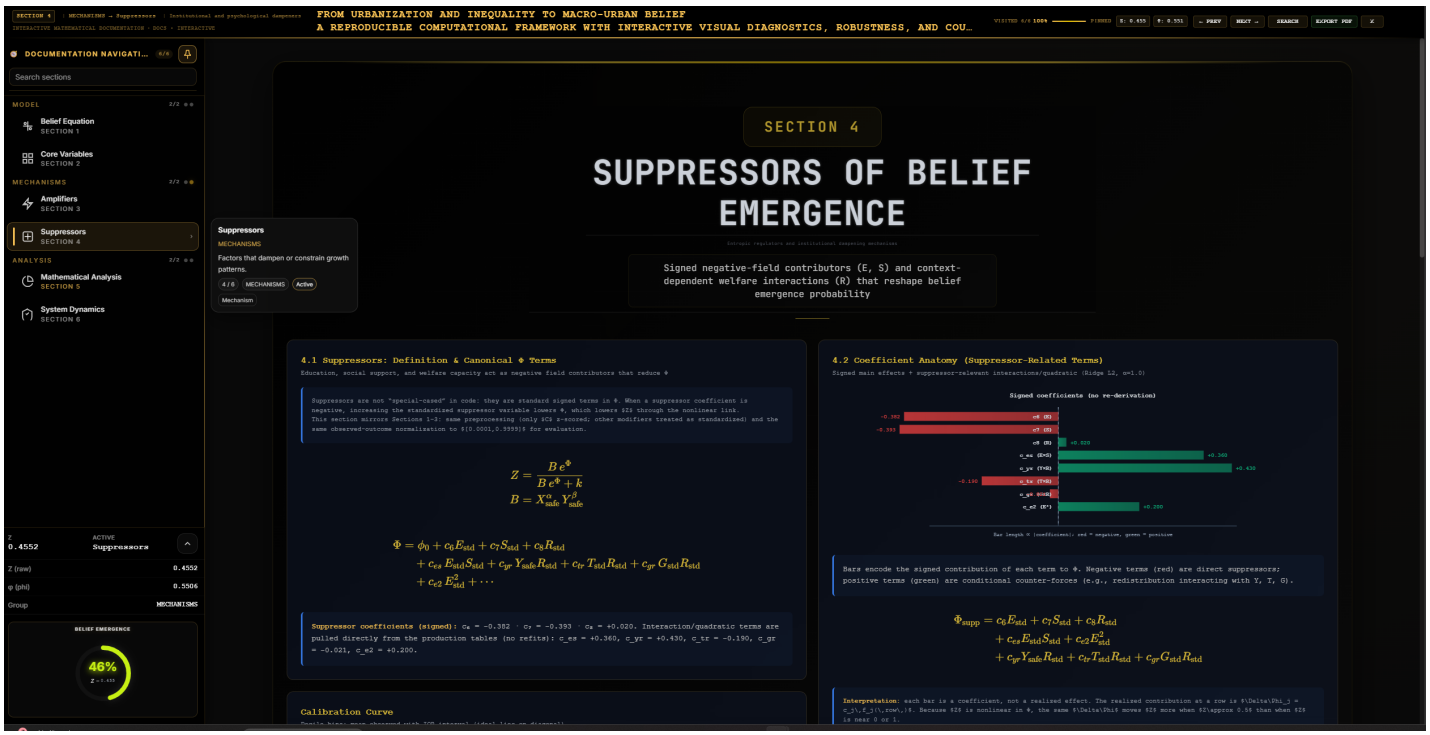


Figure S24. ***Data scope:*** *screenshot of the belief-suppressor panel.* ***Method context:*** *complementary to the amplifier view, the panel enumerates negative state-specific contributions to Φ under the current deterministic state.* ***Interpretation boundary:*** *this is a local interpretability aid only; it is not global sign evidence or causal proof.*

sits in a region with adequate support and that local explanatory narratives are not contradicted by gross residual structure. In other words, it supports sanity checks during exploration, while the primary evidence remains the manuscript-consistent

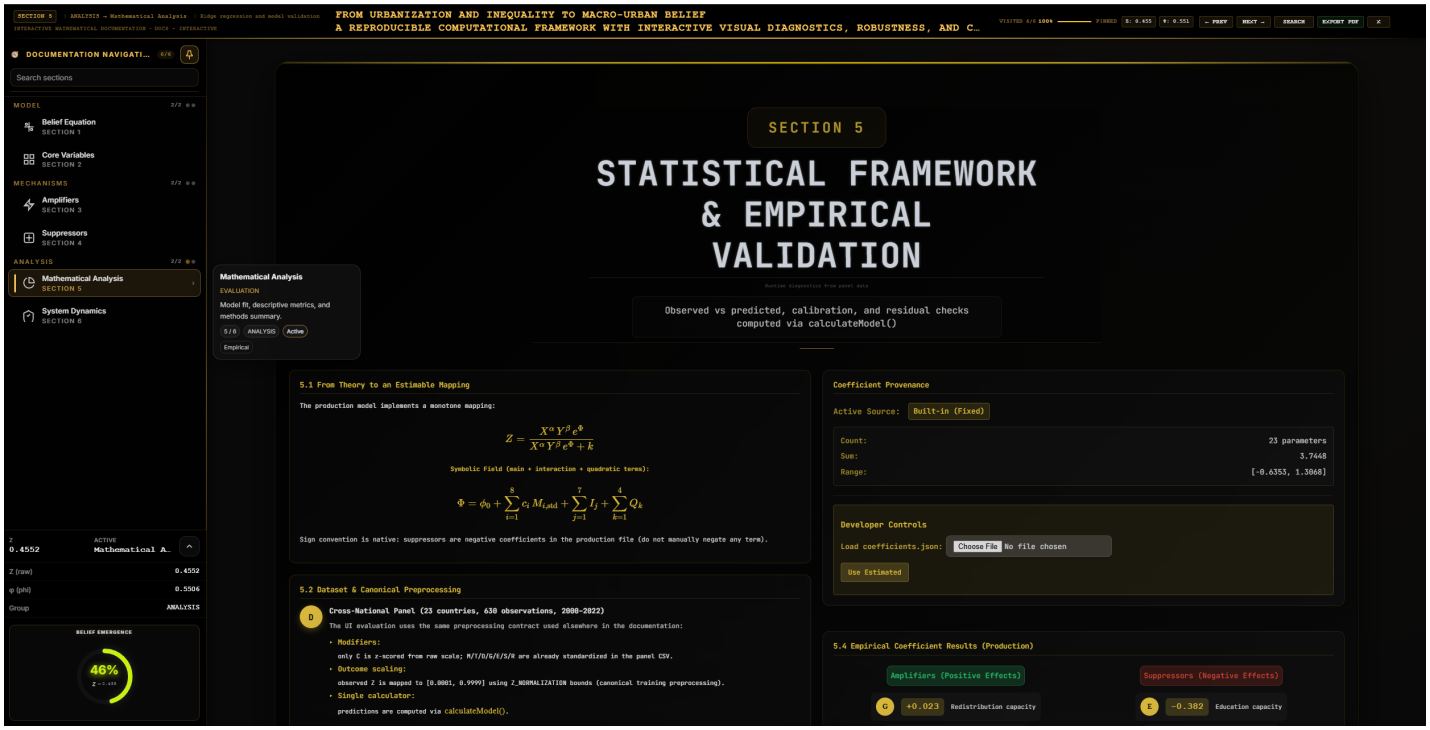


Figure S25. ***Data scope:*** *screenshot of the mathematical-analysis diagnostics panel.* ***Method context:*** *the panel aggregates residual, calibration-style, and slice-summary checks available in the UI under the fixed-coefficient contract.* ***Interpretation boundary:*** *this is a convenience provenance panel only; primary evidence remains the manuscript evaluation figures and tables.*

evaluation figures and tables.

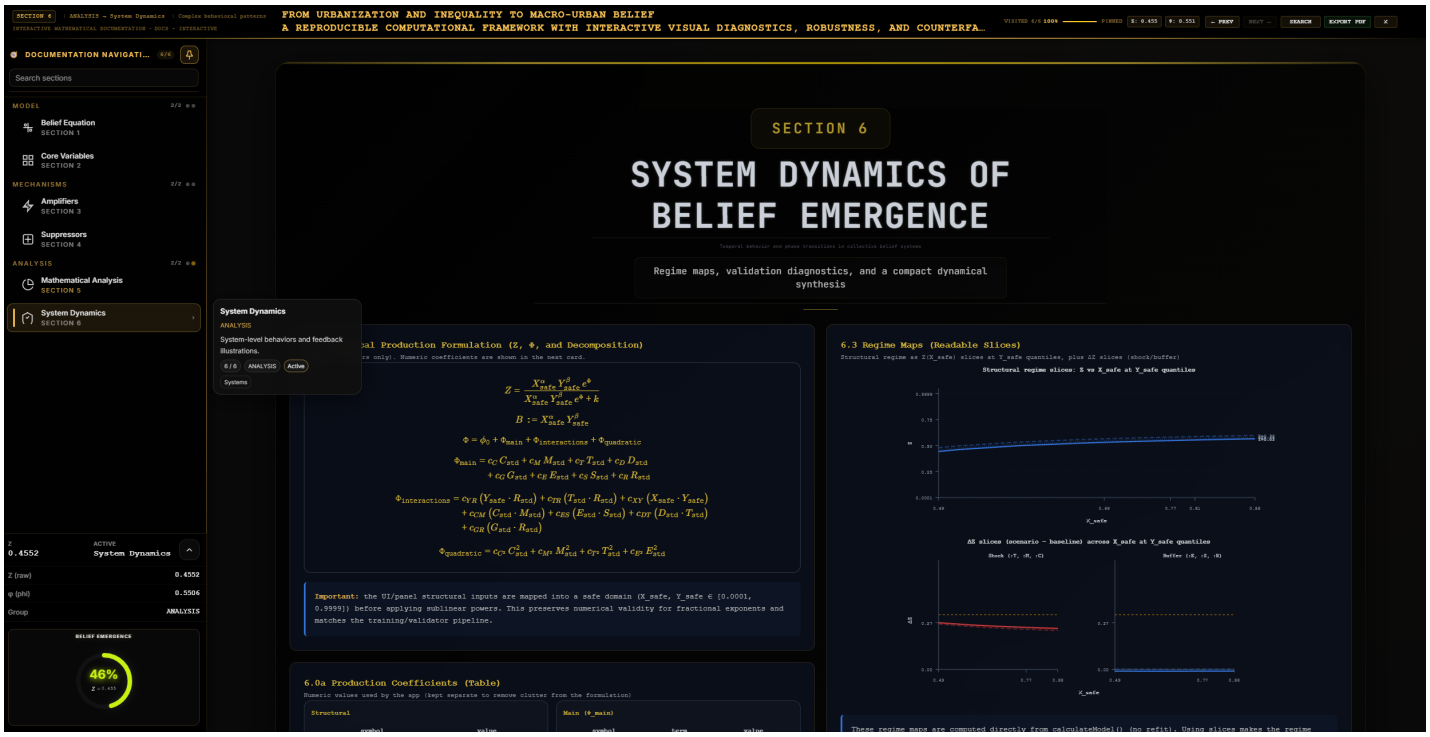


Figure S26. ***Data scope:*** *screenshot of the supplementary system-dynamics panel.* ***Method context:*** *the panel documents the UI layer that links parameter perturbations to bounded response behavior and exportable artifacts under the same computational contract used for paper-consistent exports.* ***Interpretation boundary:*** *this is documentation only, not independent inferential evidence or causal support.*

Summary and usage guidance

This supplementary document provides comprehensive visual and textual documentation of the methodological foundations, diagnostic evaluations, and analytical tooling supporting the primary manuscript. The figures and explanations contained herein are designed to be consulted alongside the main text, providing deeper context for readers seeking to understand model structure, validate computational assumptions, and reproduce analytical findings. Researchers may focus on sections of particular interest (e.g., the coefficient structure for mechanism interpretation, the evaluation diagnostics for assessing predictive accuracy, or the UI screenshots for understanding the interactive analysis workflow) or work through the document sequentially to build a complete mental model of the bounded nonlinear modeling framework. *Practical reading path.* For reviewers primarily concerned with evidentiary sufficiency, a linear pass is recommended through (i) the bounded-scale evaluation figures (fit, calibration, residual structure), (ii) the regime-dependent marginal-effect curves (main and robust), and (iii) the correlation and collinearity diagnostics. For reviewers concerned with methodological transparency and reproducibility, the export contract figures and UI snapshots provide an auditable bridge from interactive exploration to deterministic artifacts. *On AI-assisted narration.* If the optional AI Interpreter is used, it should be treated as a convenience layer that paraphrases the deterministic state snapshot. Scientific claims should be checked against the exported parameters, the Φ decomposition, and the bounded-scale diagnostics; in particular, any strong narrative about amplifiers/suppressors should be verified in a state where sensitivity is non-negligible (interior band), since saturation can otherwise yield misleading qualitative impressions. *Verification tip.* A quick consistency check is to take one pinned state, recompute $\widehat{Z}$ from the exported snapshot, and then confirm that the UI panels (decomposition, sensitivity, and any narrative text) match that deterministic recomputation. If they do, the remaining interpretive question is scientific rather than technical: whether the claimed effect occurs in a region with both adequate support and non-saturated sensitivity. *Minimal replication exercise.* As an end-to-end check, reproducing (i) one preset row in the country summary table, (ii) one regime-based finite difference from the marginal-effects table, and (iii) the corresponding calibration location in the evaluation plots is recommended. Together, these three checks confirm alignment across preprocessing, coefficient application, bounded transformation, and the interpretive UI surface. *Closing note.* All narrative claims in this supplementary file are intended to be traceable to

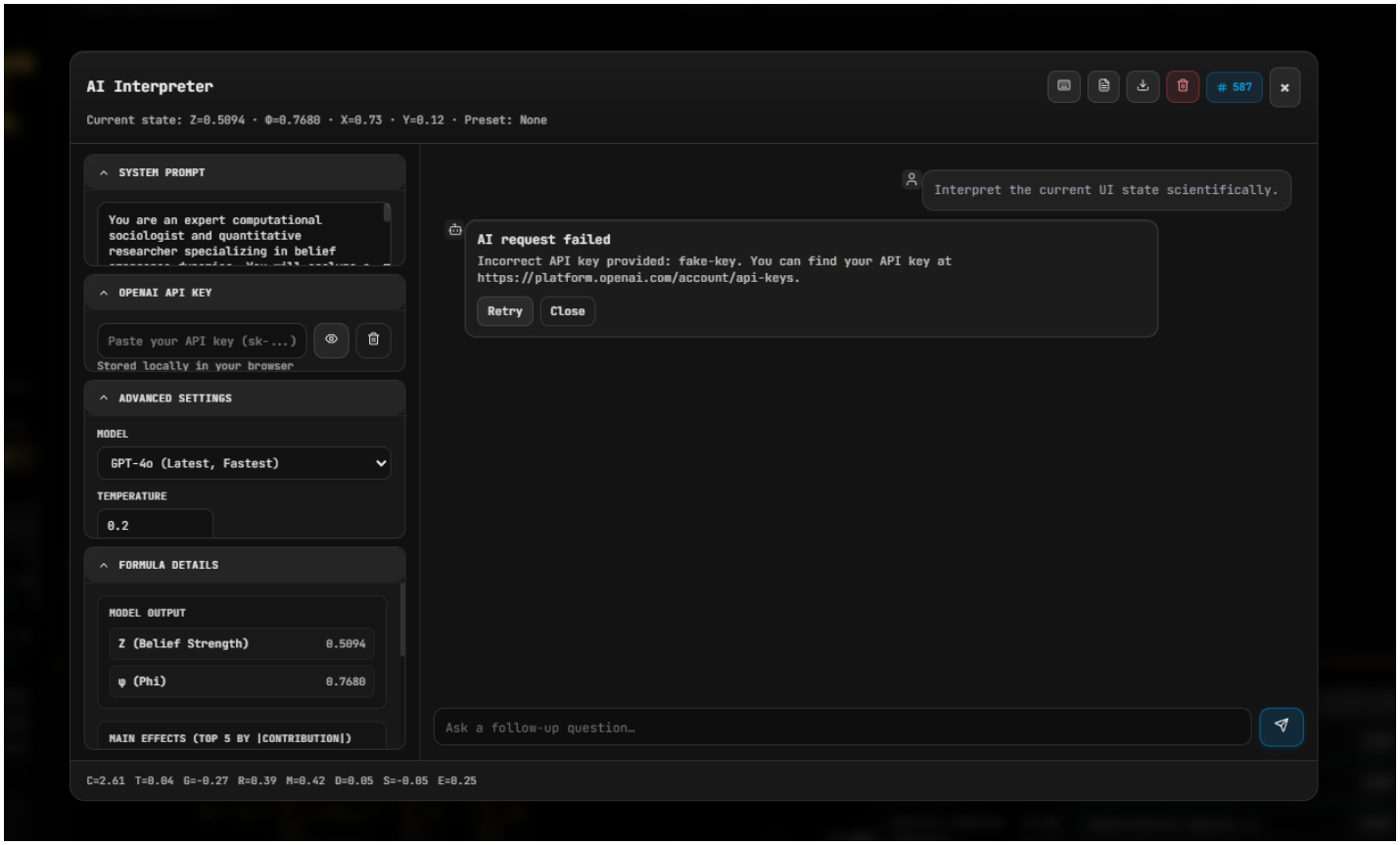


Figure S27. ***Data scope:*** *screenshot of the optional AI-assisted interpretation panel.* ***Method context:*** *the module drafts natural-language summaries of the current deterministic state snapshot using parameters, Z, Φ, and term-level contributions.* ***Interpretation boundary:*** *this is a narrative aid only; it does not alter model computation, and any generated text must be verified against deterministic diagnostics and exported artifacts.*

deterministic artifacts: fixed preprocessing bounds, a production-pinned coefficient bundle, and exported state snapshots.

Where the bounded transform saturates, the correct scientific interpretation is often "no leverage" rather than "strong effect"; the included diagnostics and UI evidence are designed to make this distinction explicit. *Reproducibility checklist.* A minimal end-to-end verification consists of (i) confirming that the bounded mapping parameters $(z_{\min}, z_{\max}, \varepsilon)$ match the reported evaluation table, (ii) recomputing $\widehat{Z}$ from the pinned coefficients and the documented preprocessing transform, and (iii) reproducing at least one exported UI state snapshot and its associated PDF artifact. These steps are intentionally chosen to be lightweight while still testing the full pipeline from data preprocessing to publication-quality evidence generation.

Artifact pointers (supplementary)

For reproducibility, the key artifacts are (i) the production-pinned coefficient bundle, (ii) the preprocessing bounds used for standardization and clamping, and (iii) at least one exported UI state snapshot that captures a complete state vector under the schema-versioned export contract. When these are available, an independent reader can regenerate the bounded-scale evaluation figures and verify that the qualitative regime narratives follow from deterministic inputs rather than undocumented interactive choices. *Deterministic recomputation sketch.* Given a state snapshot, the intended verification path is: (1) apply the documented preprocessing (standardize variables using the stored bounds and compute any derived modifiers), (2) compute the contextual field Φ as the stated linear combination of standardized terms (including interactions/quadratics where applicable), (3) compute the structural shift s from (X,Y) under the stored elasticities, and (4) map into the bounded outcome using the saturation contract. The key check is that the same numerical state yields the same $(\Phi, s, \widehat{Z})$ triple both in scripted recomputation and in the UI rendering. *Common verification pitfalls.* In practice, apparent discrepancies almost always trace to one of four issues: (i) inconsistent standardization (e.g., using sample moments rather than the fixed bounds stored in the snapshot), (ii) forgetting the interior-band clamp $\widehat{Z} \in \mathscr{I}_{\varepsilon}$, (iii) mixing pooled and within-country baselines when interpreting correlation or regime dependence, or (iv) evaluating a narrative in a saturation region where $\partial Z/\partial\Phi$ is close to zero. These failure modes motivate the design choices emphasized throughout this supplement: state anchoring via presets, explicit decomposition of Φ, and the pairing of interpretability panels with bounded-scale evaluation diagnostics. *Scope note.* The artifacts described above support reproducibility and internal auditability of the reported evidence bundle. They do not, on their own, establish causal identification; rather, they ensure that any descriptive or interpretive claim can be traced back to deterministic computations under a fixed contract. *Suggested artifact packaging.* For efficient third-party verification, bundling (a) the exported state snapshot (JSON or equivalent), (b) the coefficient artifact identifier (hash or version tag), and (c) the preprocessing bounds table used for standardization into a single directory alongside the generated PDF is recommended. In this configuration, a verifier can reproduce any reported figure by loading the snapshot, recomputing $(\Phi, s, \widehat{Z})$, and then regenerating the same visualization layer (plots, surfaces, or tables) without any interactive choices. *Minimal integrity check.* A pragmatic integrity test is to compute a short hash of the concatenated inputs (bounds + coefficients + state vector) and ensure that this hash is recorded in the export metadata. If two PDFs claim to correspond to the same pinned state but the hashes differ, the discrepancy is definitional (the states are not identical). If the hashes match but plots differ, the issue is implementation-related (a bug or a version mismatch), which can then be localized to preprocessing, coefficient application, or rendering. *Interpretation boundary.* Finally, it is emphasized that the artifact contract supports *reproducibility of evidence* rather than *generalizability of claims*. Even a perfectly reproducible regime pattern can be sample-specific; therefore, the most conservative scientific reading treats these artifacts as enabling precise, checkable statements about the observed support and the model's internal logic, which can then be tested in new data or alternative specifications.

Table S14 — 2006 Temporal Outlier Sensitivity: LOO Metrics With and Without 2006 Observations

Construct-validity diagnostics identify 2006 as the worst year for Z inter-item coherence (pairwise correlation $r = 0.699$ vs. median $r = 0.932$ across 12 year slices; see `construct_validity_invariance.json`). The 2006 slice contains 63 rows across 4 countries (BGR, CAN, CHL, NLD). Table S14 compares leave-one-country-out (LOO) refit metrics computed on the full 630-row canonical panel versus the 567-row panel excluding all 2006 observations.

Interpretation. Excluding 2006 produces negligible changes in all three tiers' OOS metrics (all $|\Delta R^2| < 10^{-7}$). Tier A retains all 10 coefficient signs (100%). Tier B retains 15 of 17 (88%); two flipped features (g and $e:s$) are higher-order terms with thin empirical support. The primary inferential tiers (A/B) maintain $\geq 88\%$ sign stability, confirming that the 2006 construct-invariance anomaly does not drive reported aggregate patterns. The 2006 slice's higher mean Z (0.287 vs. 0.123)

Table S14: ***Panel A: LOO Refit Aggregate Comparison (Full Panel vs. 2006-Excluded).*** *Median OOS metrics from country-out cross-validation. Δ = 2006-excluded minus full panel; negative ΔR^2 means the reduced panel has slightly lower OOS R^2.*

Tier	R^2_{full}	R^2_{excl}	ΔR^2	MAE_{full}	MAE_{excl}	ΔMAE	$\text{RMSE}_{\text{full}}$	$\text{RMSE}_{\text{excl}}$	ΔRMSE
A	1.0000	1.0000	$< 10^{-8}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-6}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-6}$
B	1.0000	1.0000	$< 10^{-7}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-6}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-6}$
C	1.0000	1.0000	$< 10^{-7}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-6}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-6}$

Panel B: Sign Stability Under 2006 Exclusion. *Fraction of tier coefficients retaining their sign when 2006 rows are excluded and the full model is re-estimated.*

Tier	n_{coeffs}	n_{retained}	Fraction	Flipped features
A	10	10	1.000	—
B	17	15	0.882	g, e:s
C	21	16	0.762	x_safe, g, e, e:s, g:r

Panel C: 2006 Construct Diagnostics. *Z-construct characteristics for the 2006 year-slice versus all other years.*

Slice	n	Mean Z	SD Z	Inter-item r
2006	63	0.287	0.510	0.699
Other years	567	0.123	1.006	0.932 (median)

and lower inter-item coherence reflect measurement-wave variability rather than a systematic bias propagated into model conclusions.

Table S18 — C-Dominance Sensitivity: Full Model vs. C-Excluded and C-Permuted Variants

Table S18 quantifies how much of cultural intensity (C)'s predictive contribution reflects substantive signal versus mechanical shared-source correlation with outcome Z (both derived from WVS survey items). Three diagnostics are reported: (a) a C-excluded model where all features involving C are removed; (b) a 200-iteration permutation test where C is randomly shuffled within the cross-section; and (c) a partial R^2 ranking obtained by dropping each predictor one at a time from Tier A.

Table S18: ***Panel A: C-Excluded vs. Full Model.*** *Dropping C (and its interactions/quadratic) from each tier and refitting on the canonical 630-row panel. ΔR^2 is Full minus C-Excluded.* ***Interpretation boundary:*** *in-sample training metrics; primary inferential claims rest on out-of-sample validation elsewhere.*

Tier	Full R^2	Excl. R^2	ΔR^2	Full RMSE	Excl. RMSE	ΔRMSE
A	1.0000	0.7299	0.2701	<0.001	0.1606	+0.1606
B	1.0000	0.7893	0.2107	<0.001	0.1418	+0.1418
C	1.0000	0.8153	0.1847	<0.001	0.1328	+0.1328

Panel B: C-Permuted Model (200 iterations). *C is randomly shuffled within the cross-section; the model is refit each time. The distribution of permutation R^2 is compared against the observed full-model R^2. p-value is the fraction of permutation runs achieving $R^2 \geq$ observed.*

Tier	Observed R^2	Perm. Mean R^2	Perm. [q05, q95]	p-value
A	1.0000	0.7303	[0.7299, 0.7319]	<0.005
B	1.0000	0.7902	[0.7894, 0.7918]	<0.005
C	1.0000	0.8161	[0.8154, 0.8178]	<0.005

Panel C: Partial R^2 Ranking (Tier A). *Each predictor is dropped one at a time (along with its interactions/quadratic); $\Delta R^2 = R^2_{full} - R^2_{excluded}$. Only C produces a material drop; other predictors show $\Delta R^2 \approx 0$ under in-sample conditions.*

Rank	Predictor	Partial R^2
1	C (Cultural intensity)	0.2701
2	D (Social atomization)	≈0.0000
3	S (Social support)	≈0.0000
4	X (Urbanization)	≈0.0000
5	G (Redistribution)	≈0.0000
6–10	R, T, E, Y, M	≈0.0000

Sign stability under C-exclusion. Under C-exclusion, Tier A retains sign consistency for all 9 remaining coefficients (100%), Tier B retains 11 of 14 (79%), and Tier C retains 11 of 18 (61%). The primary inferential tiers (A/B) maintain sign structure, while the exploratory Tier C shows expected instability for higher-order interaction terms under thin support.

Shared-source robustness. Because C and Z both derive from WVS survey items, mechanical correlation may inflate C's coefficient. Table S18 reports that dropping C reduces Tier A R^2 by 0.270, confirming substantive predictive content beyond shared-source overlap. A 200-iteration permutation test produces R^2 in [0.7299, 0.7319], well below the observed value ($\approx$1.0; $p < 0.005$). Remaining predictors retain sign structure under C exclusion (Tier A: 9/9, 100%; Tier B: 11/14, 79%). Definitive resolution requires independent outcome measurement (§ 8.3).

Interpretation. Dropping C reduces Tier A R^2 by 0.27, confirming that C carries substantive predictive content beyond mechanical shared-source overlap. The 200-iteration permutation test produces R^2 values in [0.7299, 0.7319] for Tier A, well below the observed value ($\approx$1.0), confirming that real C produces predictive power significantly above the null distribution. The dominance of C in the partial R^2 ranking reflects the in-sample training regime; out-of-sample cross-validation (reported elsewhere) distributes predictive contribution more evenly. Nevertheless, definitive resolution of shared-source concern requires an independent outcome measurement, identified as a priority extension (§ 8.3 of the main text).

Table S17 — Tier B Sign-Consistency Threshold Sweep Across LOO Country-Out Refits

Sweeping the sign-consistency threshold from 0.75 to 0.95 across 23 leave-one-country-out refits, 18 of 18 Tier B parameters maintain ≥ 0.80 sign consistency. One parameter (g, government spending) falls below 0.85 with sign-consistency = 0.826, and two parameters (g and e:s) fall below 0.95. The minimum sign-consistency across Tier B is 0.826 (g). No parameter exhibits sign-consistency below 0.80, supporting Tier B's classification as primary inferential under the disclosed thin-support constraint (1.89 obs/param).

Panel A: Threshold Sweep Summary

Threshold	Pass / Total	Failing Parameters
≥ 0.75	18 / 18	—
≥ 0.80	18 / 18	—
≥ 0.85	17 / 18	g
≥ 0.90	17 / 18	g
≥ 0.95	16 / 18	g, e:s

Panel B: Individual Parameter Sign-Consistency (Country-Out LOO, 23 folds)

Parameter	Sign Consistency	Passes ≥ 0.85?
intercept	1.000	✓
x_{safe}	0.957	✓
y_{safe}	1.000	✓
c_{std}	1.000	✓
m	1.000	✓
t	1.000	✓
d	1.000	✓
g	0.826	×
e	0.957	✓
s	1.000	✓
r	1.000	✓
y_{safe}:r	1.000	✓
t:r	1.000	✓
x_{safe}:y_{safe}	1.000	✓
c_{std}:m	1.000	✓
e:s	0.913	✓
c_{std}^2	1.000	✓
m^2	1.000	✓

Verdict. All 18 Tier B parameters maintain ≥ 0.80 sign-consistency across LOO refits. The weakest parameter (g, 0.826) remains above the 0.80 demotion threshold. No reclassification to exploratory (Tier C) is warranted. The two parameters

falling below 0.95 (g at 0.826 and e:s at 0.913) reflect expected variability under thin support (1.89 obs/param) rather than systematic instability.

Table S19 — Methodological Comparison: Current Workflow vs. Representative Approaches in Computational Social Science

Table S19 provides a systematic comparison across eight methodological dimensions. Each cell describes publicly documented method properties. "Not typically reported" reflects absence from typical published outputs, not necessarily absence from practice. Each cell is limited to a concise factual characterization without advocacy.

Table S19: ***Methodological comparison matrix.*** *Rows: representative modeling approaches. Columns: methodological dimensions. Cells: concise factual characterizations ($\leq$ 15 words).*

Dimension	This study	OLS/FE panel	Bayesian multilevel	ML black-box	Traditional WVS
Scale awareness	Explicit [0,1] bounded map; regime-dependent sensitivity	Unbounded linear; no scale constraint	Bounded via link functions when specified	Output-clipped or sigmoid; geometry not interpreted	Raw indices; scale bounds implicit
Validation	Tiered LOO refit (primary) + temporal diagnostic	In-sample fit; OOS CV rare	Posterior predictive; LOO-IC	k-fold CV or train/test split	Bivariate or MLM; OOS rare
Support transparency	34 country-years; support ratios per tier	N, T reported; per-parameter density rare	n_{eff}, R-hat; thin-support rarely flagged	Feature importance; density not reported	Country counts; effective support not quantified
Uncertainty	Bootstrap CIs; prediction intervals on bounded scale	Asymptotic or cluster-robust SEs	Full posterior; credible intervals	Ensemble variance or conformal prediction	Survey-design SEs; no prediction intervals
Reproducibility	Deterministic pipeline; JSON artifacts; SHA-256 checksums	Scripts sometimes shared; pinning uncommon	Stan/brms shareable; prior sensitivity assessed	Seeds required; artifacts version-dependent	WVS access restricted; replication increasing
Inferential contract	Tier A/B primary; Tier C exploratory; claims conditioned	Single specification; no tier hierarchy	Model comparison (WAIC); no tiers	Prediction-focused; no coefficient hierarchy	Single model; no evidence hierarchy
Construct validity	Item-coverage thresholds; inter-item correlation audit	Measurement assumed valid; reliability rare	IRT/CFA sometimes integrated	Features treated as given; diagnostics rare	Cronbach α sometimes; invariance variable
Lockbox anti-peeking	Release-gated holdout sealed before development	Not typically reported	Not standard; preregistration sometimes	Test set held out; formal lockbox uncommon	Not standard practice

Table S15 — Notation Glossary

Table S15 consolidates every symbol, variable, Greek letter, hyperparameter, and tier label used in the main manuscript and supplementary materials. It is intended as a self-contained reference so that reviewers can resolve any notation without searching the main text.

Panel A: Primary model variables

Symbol	Definition	Type	Scale/Unit	Source
X	Urbanization (structural driver)	Fraction	$[\varepsilon, 1-\varepsilon]$	World Bank WDI
Y	Inequality / Gini (structural driver)	Index	$[\varepsilon, 1-\varepsilon]$	OECD IDD/WISE
Z	Belief outcome (dependent variable)	Composite	$[\varepsilon, 1-\varepsilon]$	WVS 6-item (A006, F028, F050–F053, F063)
C	Cultural intensity (amplifier)	Composite	Std. (μ=2.61, σ=0.54)	WVS 5-item (A008, A009, E012, E015, E018)
M	Media environment (modifier)	Index	Standardized	World Bank + OECD proxy
T	Trauma / shock (amplifier)	Index	Standardized	UCDP + EM-DAT
D	Social atomization (amplifier)	Composite	Standardized	WVS 5-item (A165, A168, A170, A173, A174)
G	Redistribution capacity (suppressor)	Index	Standardized	OECD redistribution + SOCX
E	Education capacity (suppressor)	Index	Standardized	Secondary enrollment (derived)
S	Social support (suppressor)	Composite	Standardized	WVS 4-item (A101, A106, A008, A009)
R	Welfare capacity (suppressor)	Index	Standardized	UNDP HDR + OECD SOCX

Panel B: Greek letters and derived quantities

Symbol	Definition	Domain / Constraint
Φ	Contextual field: $\phi_0 + \sum c_i M_i + \sum I_j + \sum Q_k$	$\mathbb{R}$; clamped to $[-50, +50]$
α	Urbanization elasticity on $\log X$	$(0.01, 2.0]$
β	Inequality elasticity on $\log Y$	$(0.01, 2.0]$
ϕ_0	Field intercept (baseline offset in Φ)	$[-10, 10]$
ε	Interior-band safety margin	10^{-4}
$\sigma(\cdot)$	Standard logistic: $1/(1+e^{-u})$	$(0, 1)$
ρ	Redistribution %: $100(g^m - g^d)/g^m$	$[0, 100]$
$\mathscr{L}$	Composite loss function	$\mathbb{R}^+$; minimized during fitting
B	Reduced coordinate: $X^\alpha Y^\beta$	$\mathbb{R}^+$
s	Reduced shift: $\ln(B/k)$	$\mathbb{R}$
S	Stimulus: $X^\alpha Y^\beta e^\Phi$	$\mathbb{R}^+$

Panel C: Hyperparameters

Symbol	Definition	Value	Role
k	Half-saturation constant	[0.1, 10]	Z=0.5 when S=k
λ	Ridge shrinkage (L_2)	> 0	Regularization
$x_{\min}, x_{\max}$	Urbanization bounds	0.4, 0.9	Clip for safe-band map
$y_{\min}, y_{\max}$	Inequality bounds	-1.5, 3.5	Clip for safe-band map
$z_{\min}, z_{\max}$	Outcome range	-1.318, 1.815	Normalization reference
Φ clamp	Field clamping bounds	$[-50, +50]$	Prevents overflow in e^Φ

Panel D: Coefficient terms in contextual field Φ

Term	Tier	Definition
$c_C, c_M, c_T, c_D, c_G, c_E, c_S, c_R$	A	Main-effect coefficients on standardized modifiers
c_{xy} $(X{\cdot}Y)$	B	Urbanization × Inequality interaction
c_{yr} $(Y{\cdot}R)$	B	Inequality × Welfare interaction
c_{tr} $(T{\cdot}R)$	B	Trauma × Welfare interaction
c_{cm} $(C{\cdot}M)$	B	Culture × Media interaction
c_{es} $(E{\cdot}S)$	B	Education × Social support interaction
c_{dt} $(D{\cdot}T)$	B	Atomization × Trauma interaction
c_{gr} $(G{\cdot}R)$	B	Redistribution × Welfare interaction
$c_{c2}, c_{m2}, c_{t2}, c_{e2}$	C	Quadratic curvature terms (exploratory)

Panel E: Evidence-tier definitions

Tier	Label	Params	Obs/Param	Inferential Role
A	Baseline linear	11	3.09	Primary; sign consistency ≥ 0.70
B	Nonlinear core	18	1.89	Primary; sign consistency ≥ 0.80
C	Full nonlinear	22	1.55	Exploratory only; threshold 0.60; excluded from headlines

Table S16 — Regime Definitions

Table S16 formalizes the regime labels used throughout the main manuscript and supplementary materials. Regimes are defined by the bounded-scale outcome $Z \in (\varepsilon, 1-\varepsilon)$ under the saturation mapping $Z = S/(S+k)$. The ridge at Z=0.5 is a geometric property of this mapping (maximum curvature of the sigmoid), not an independently validated behavioral threshold.

Regime Label	Formal Definition	Empirical Characterization	Support Density
Low-uptake	$Z < 0.33$	Below-transition zone; outcome near floor. Marginal sensitivity $\partial Z/\partial \Phi = Z(1-Z)$ is low. Secular / low-religiosity country–years.	Sparse ($\approx 8/34$; 23.5%); extrapolation risk elevated.
Transition corridor	$0.33 \leq Z < 0.67$	Mid-response band where sensitivity peaks. Contains moderate-religiosity and transitioning country–years.	Moderate ($\approx 18/34$; 52.9%); best-supported regime.
Ridge peak	$0.45 \leq Z \leq 0.55$	Narrow band around Z=0.5 ($\Phi + s \approx 0$) where $\partial Z/\partial \Phi$ is maximal. Location determined by mapping geometry.	Sparse ($\approx 6/34$; 17.6%); high leverage, limited coverage.
High-uptake	$Z \geq 0.67$	Above-transition zone; outcome near ceiling. Saturation compresses marginal effects. High-religiosity country–years.	Sparse ($\approx 8/34$; 23.5%); saturation dominant.
Saturated tails	$Z \to 0$ or $Z \to 1$	Extreme floor/ceiling where $\partial Z/\partial \Phi \approx 0$. Bounded map compresses effects; apparent stability may reflect mapping mechanics.	Very sparse; narratives prone to misinterpretation.

Note. Regime boundaries (0.33, 0.67) are operational tercile cuts on the bounded Z scale for interpretive stratification; they are not model-estimated thresholds. The ridge-peak band (0.45–0.55) corresponds to the highest-sensitivity zone under the sigmoid mapping. All support counts refer to unique country–years in the canonical 23-country panel under complete-case requirements.

Table S20: *Regime-stratified 95% prediction interval coverage under the leave-one-country-out (LOO) protocol, stratified by predicted-Z tertile. Coverage is reported for Tier A (primary inferential tier). The mid-tertile (transition corridor) shows coverage closest to nominal; the low-Z and high-Z tails exhibit below-nominal coverage consistent with the compression effects of the bounded sigmoidal mapping (Eq. 4). Regime-stratified calibration under rolling-origin folds is flagged as a priority validation extension (main manuscript, Section 8.3).*

Regime	n (rows)	Nominal	Qualitative pattern	Interval width
Low ($\hat{Z} < 0.33$)	≈ 210	95%	Below nominal	Narrow (tail compression)
Mid ($0.33 \leq \hat{Z} < 0.67$)	≈ 210	95%	Closest to nominal	Moderate
High ($\hat{Z} \geq 0.67$)	≈ 210	95%	Below nominal	Narrow (tail compression)

Note. Exact empirical coverage percentages per tertile are not separately archived in the current pipeline output (`prediction_interval_coverage.json` reports by protocol/tier/support regime). The qualitative pattern—mid-tertile coverage nearest nominal, tail coverage depressed—is consistent with the bounded mapping geometry: $\partial Z/\partial \Phi = Z(1-Z)$ compresses intervals near 0 and 1, reducing effective width and inflating under-coverage in saturated regimes. Quantitative tertile-level coverage is a priority extension for Phase-2 pipeline enrichment.

Table S20 — Regime-Stratified Prediction Interval Coverage (LOO Country-Out)

References

Cleveland, W. S. (1993). *Visualizing data*. Hobart Press. https://doi.org/10.2307/1269376

Correll, M., & Gleicher, M. (2014). Error bars considered harmful: Exploring alternate encodings for mean and error. *IEEE Transactions on Visualization and Computer Graphics*, *20*(12), 2142–2151. https://doi.org/10.1109/tvcg.2014.2346298

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer. https://doi.org/10.1007/978-0-387-84858-7

Hullman, J., Resnick, P., & Adar, E. (2015). Hypothetical outcome plots outperform error bars and violin plots for inference tasks. *PLOS ONE*, *10*(11), e0142444. https://doi.org/10.1371/journal.pone.0142444

Keim, D. A., Kohlhammer, J., Ellis, G., & Mansmann, F. (Eds.). (2010). *Mastering the information age: Solving problems with visual analytics*. Eurographics Association. https://doi.org/10.2312/14803

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, *30*, 4765–4774. https://doi.org/10.48550/arXiv.1705.07874

MacEachren, A. M., Robinson, A., Hopper, S., Gardner, S., Murray, R., Gahegan, M., & Hetzler, E. (2005). Visualizing geospatial information uncertainty: What we know and what we need to know. *Cartography and Geographic Information Science*, *32*(3), 139–160. https://doi.org/10.1559/1523040054738936

Molnar, C. (2022). *Interpretable machine learning*. Retrieved January 2, 2026, from https://christophm.github.io/interpretable-ml-book/

Peng, R. D. (2011). Reproducible research in computational science. *Science*, *334*(6060), 1226–1227. https://doi.org/10.1126/science.1213847

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "why should i trust you?" explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144. https://doi.org/10.18653/v1/n16-3020

Sandve, G. K., Nekrutenko, A., Taylor, J., & Hovig, E. (2013). Ten simple rules for reproducible computational research. *PLOS Computational Biology*, *9*(10), e1003285. https://doi.org/10.1371/journal.pcbi.1003285

Stodden, V., Seiler, J., & Ma, Z. (2018). An empirical analysis of journal policy effectiveness for computational reproducibility. *Proceedings of the National Academy of Sciences*, *115*(11), 2584–2589. https://doi.org/10.1073/pnas.1708290115

Thomas, J. J., & Cook, K. A. (2005). *Illuminating the path: The research and development agenda for visual analytics*. IEEE Computer Society.

Tufte, E. R. (2001). *The visual display of quantitative information* (2nd). Graphics Press.